

RAI-818: INTELLIGENT **MOBILE ROBOTICS**

Particle Filter Algorithm

- **Introduction**
- **Particle Set**
- **Importance Sampling**
- **Particle Filter Algorithm**

Text : Principles of Robot Motion – Theory, Algorithms and Implementations
H. Choset, K.M. Lynch, S. Hutchinson, G. Kantor, W. Burgard, et al.

INSTRUCTOR:

DR. KUNWAR FARAZ AHMED KHAN

Particle Filters



• Introduction

Introduction

- Represent belief by random **samples** Estimation of **non-Gaussian, nonlinear** processes
- Common Names
 - Monte Carlo filter
 - Survival of the fittest
 - Bootstrap filter
 - Particle Filter



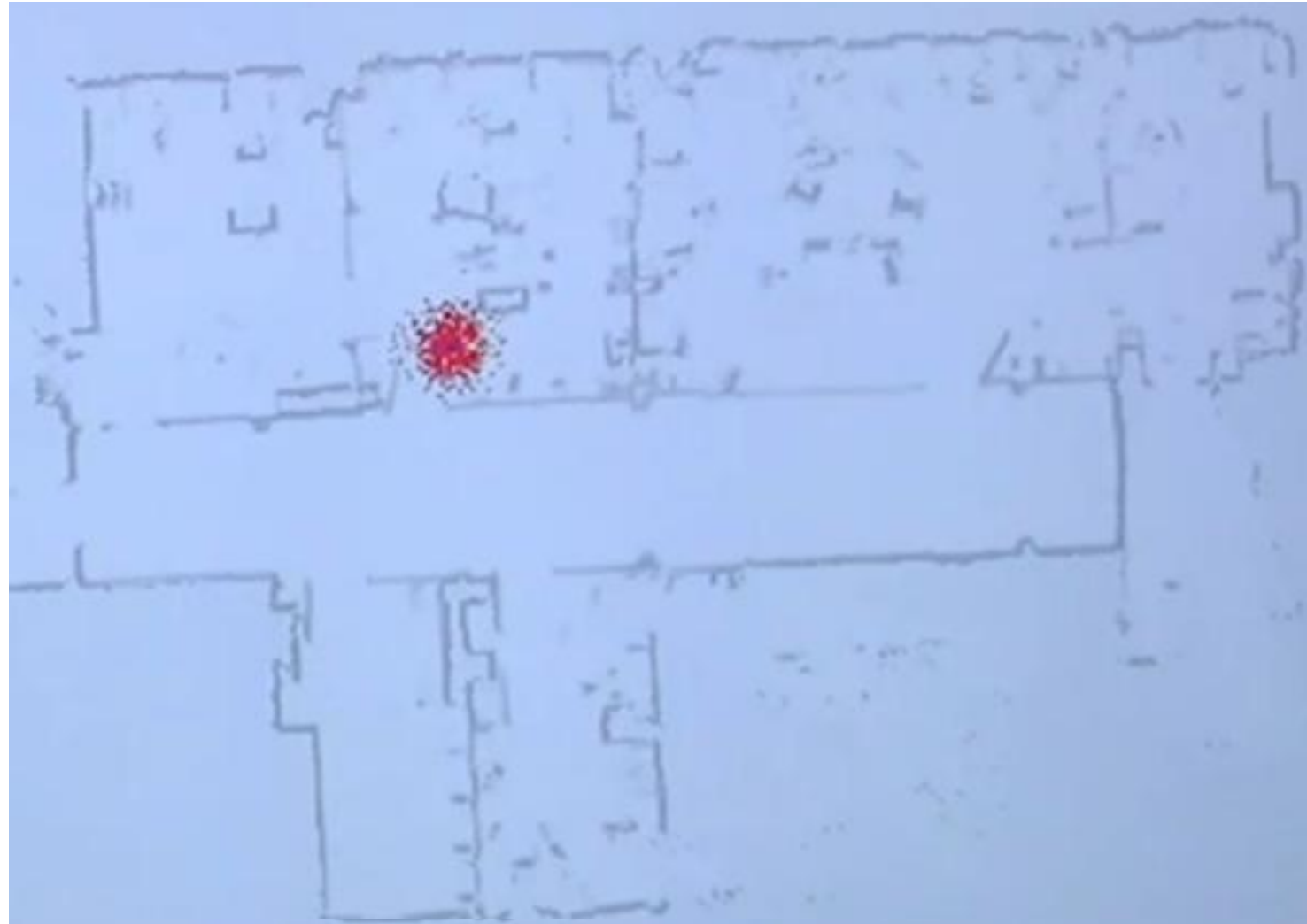
Defining futures

Particle Filters



- Introduction
 - Where am I?

Where am I?



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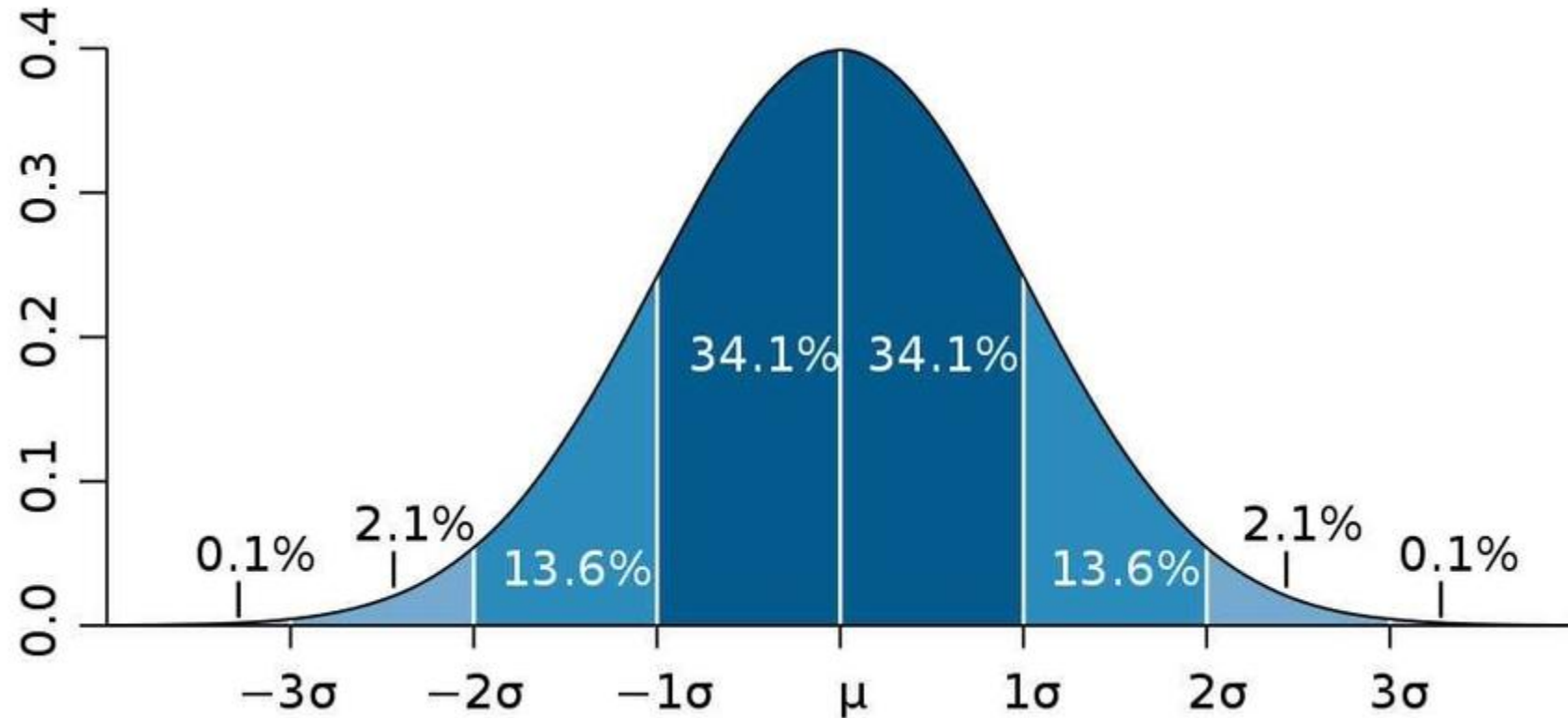


- **Introduction**
 - Where am I?
 - Gaussian Filters

Gaussian Filters

- The Kalman Filter and its variants can only model Gaussian distributions

$$p(x) = \det(2\pi\Sigma)^{-1/2} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right)$$



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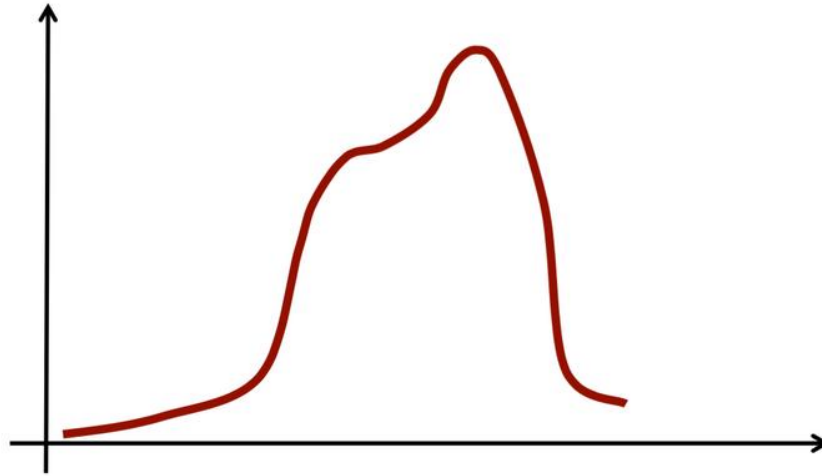


- **Introduction**

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- **Arbitrary Distributions**

Flexible Function Approximation

- Goal: approach for dealing with *arbitrary distributions*



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Particle Filters

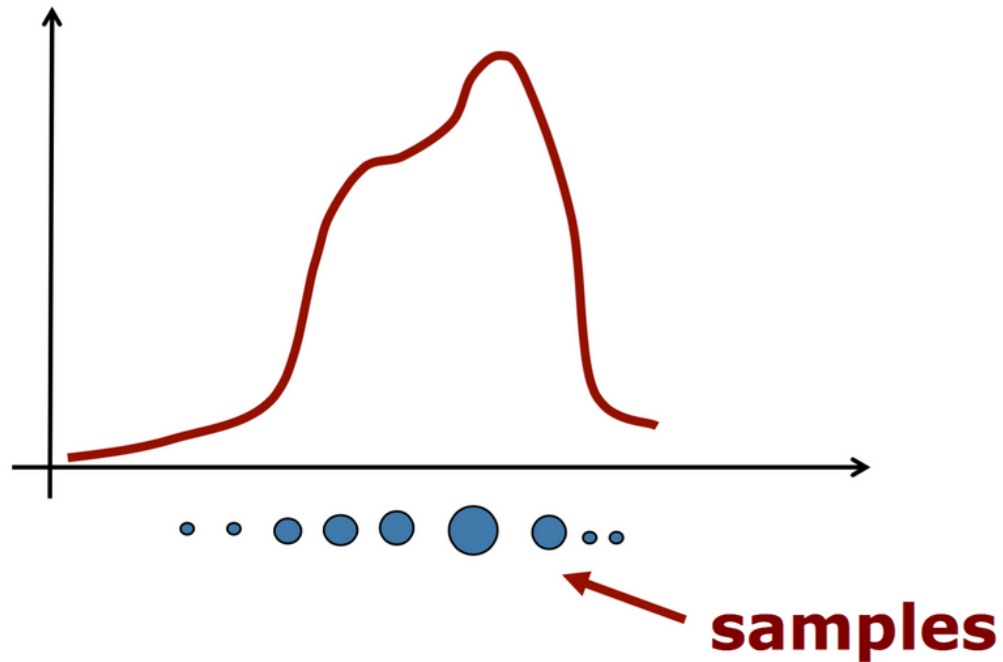


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Flexible Function Approximation

- *Multiple weighted samples* to represent *arbitrary distributions*



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Particle Filters



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Particle Set

- Set of *weighted samples*

$$\mathcal{X} = \left\{ \left\langle \underset{\substack{\text{state} \\ \text{hypothesis}}}{x^{[j]}}, w^{[j]} \right\rangle \right\}_{j=1, \dots, J}$$

state hypothesis **importance weight**

- The samples represent the posterior

$$p(x) = \sum_{j=1}^J w^{[j]} \delta_{x^{[j]}}(x)$$



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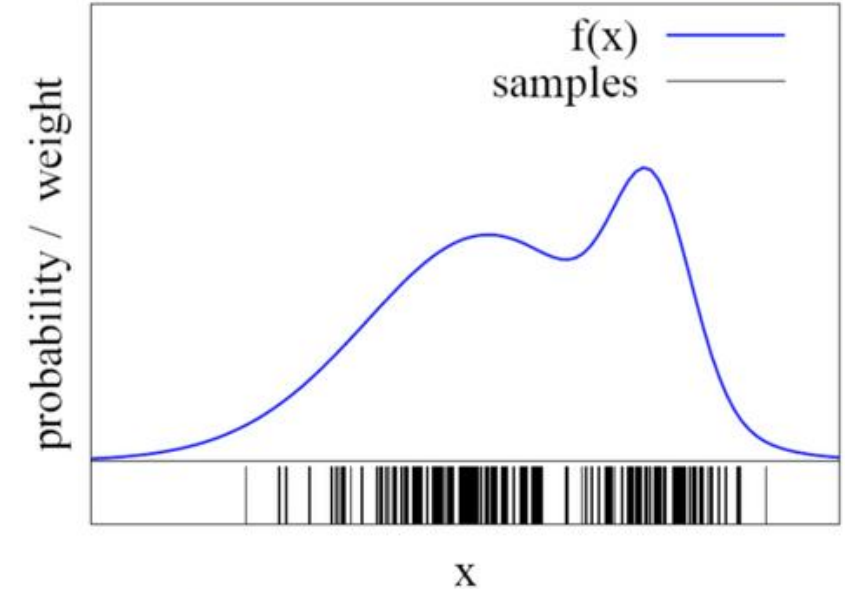
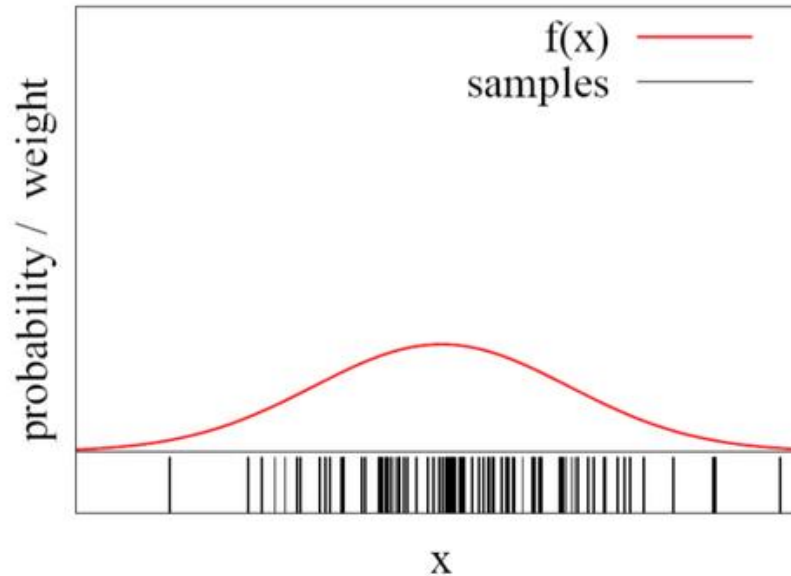


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Particle Set

- Particles for function approximation



- The more particles fall into a region, the higher the probability of the region

How to obtain such samples?



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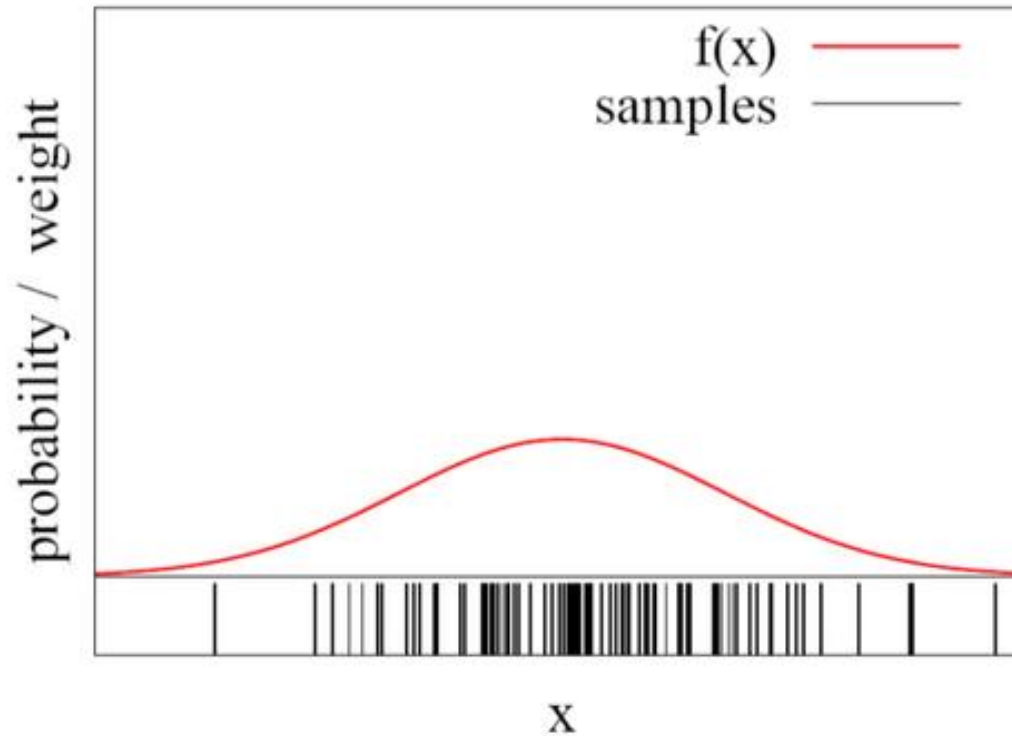
Particle Filters



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- **Particle Set**

Particle Set

- Closed form sampling is only possible for a few distributions



$$x \leftarrow \frac{1}{2} \sum_{i=1}^{12} \text{rand}(-\sigma, \sigma)$$

How to sample from other distributions?



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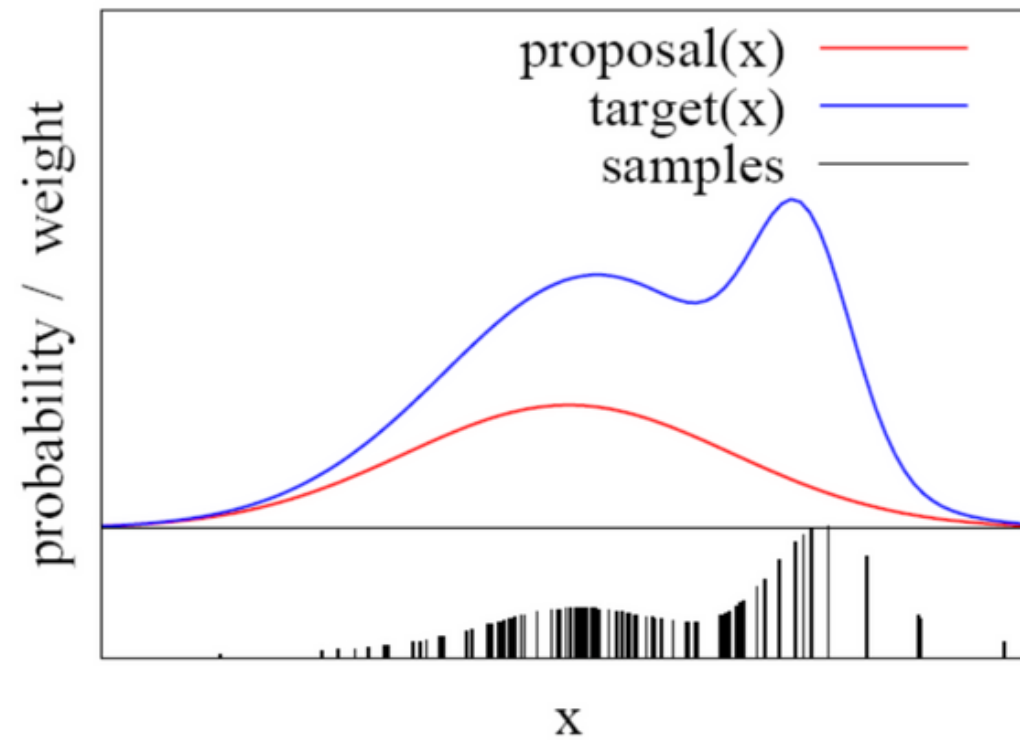
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- **Importance Sampling**

Importance Sampling Principle

- We can use a different distribution g to generate samples from f
- Account for the differences between g and f using a weight $w = f(x) / g(x)$
- target f
- proposal g
- Precondition:
 $f(x) > 0 \rightarrow g(x) > 0$



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- **Particle Filter Algorithm**

Particle Filter Algorithm

- Recursive Bayes Filter
- Non-parametric approach
- Models the distribution by samples
- Prediction: draw from the proposal
- Correction: weighting by the ratio of target and proposal

The more samples we use, the better is the estimate!



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Particle Filter Algorithm

1. Sample the particles using the proposal distribution

$$x_t^{[j]} \sim \pi(x_t \mid \dots)$$

2. Compute the importance weights

$$w_t^{[j]} = \frac{\text{target}(x_t^{[j]})}{\text{proposal}(x_t^{[j]})}$$

3. Resampling draw sample i with probability $w_t^{[i]}$ and repeat J times



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Particle Filter Algorithm

Particle_filter($\mathcal{X}_{t-1}, u_t, z_t$):

```
1:  $\bar{\mathcal{X}}_t = \mathcal{X}_t = \emptyset$ 
2: for  $j = 1$  to  $J$  do
3:   sample  $x_t^{[j]} \sim \pi(x_t)$ 
4:    $w_t^{[j]} = \frac{p(x_t^{[j]})}{\pi(x_t^{[j]})}$ 
5:    $\bar{\mathcal{X}}_t = \bar{\mathcal{X}}_t + \langle x_t^{[j]}, w_t^{[j]} \rangle$ 
6: endfor
7: for  $j = 1$  to  $J$  do
8:   draw  $i \in 1, \dots, J$  with probability  $\propto w_t^{[i]}$ 
9:   add  $x_t^{[i]}$  to  $\mathcal{X}_t$ 
10: endfor
11: return  $\mathcal{X}_t$ 
```



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Monte Carlo Localization

- Each particle is a pose hypothesis
- Proposal is the motion model

$$x_t^{[j]} \sim p(x_t \mid x_{t-1}, u_t)$$

- Correction via the observation model

$$w_t^{[j]} = \frac{\text{target}}{\text{proposal}} \propto p(z_t \mid x_t, m)$$



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4:      $w_t^{[j]} = p(z_t \mid x_t^{[j]})$ 
5:      $\bar{\mathcal{X}}_t = \bar{\mathcal{X}}_t + \langle x_t^{[j]}, w_t^{[j]} \rangle$ 
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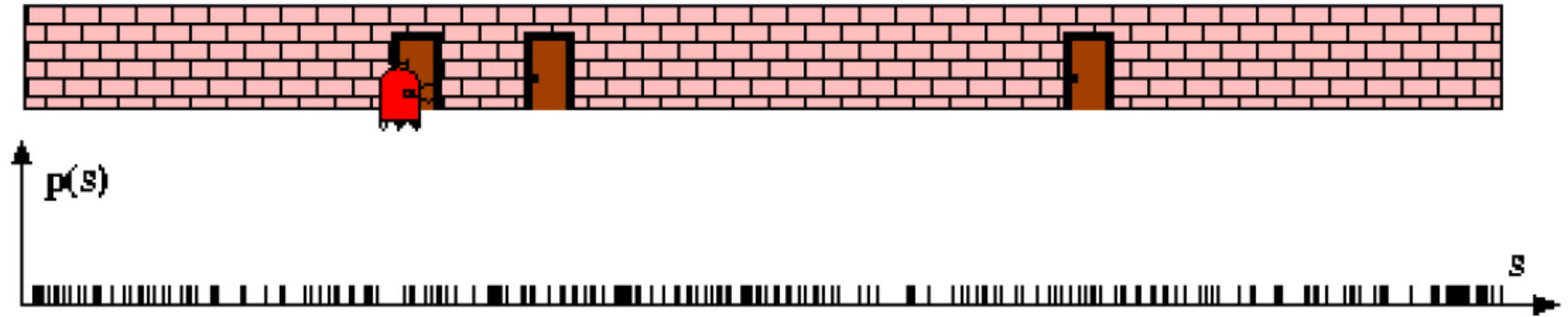


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Particle Filter Algorithm



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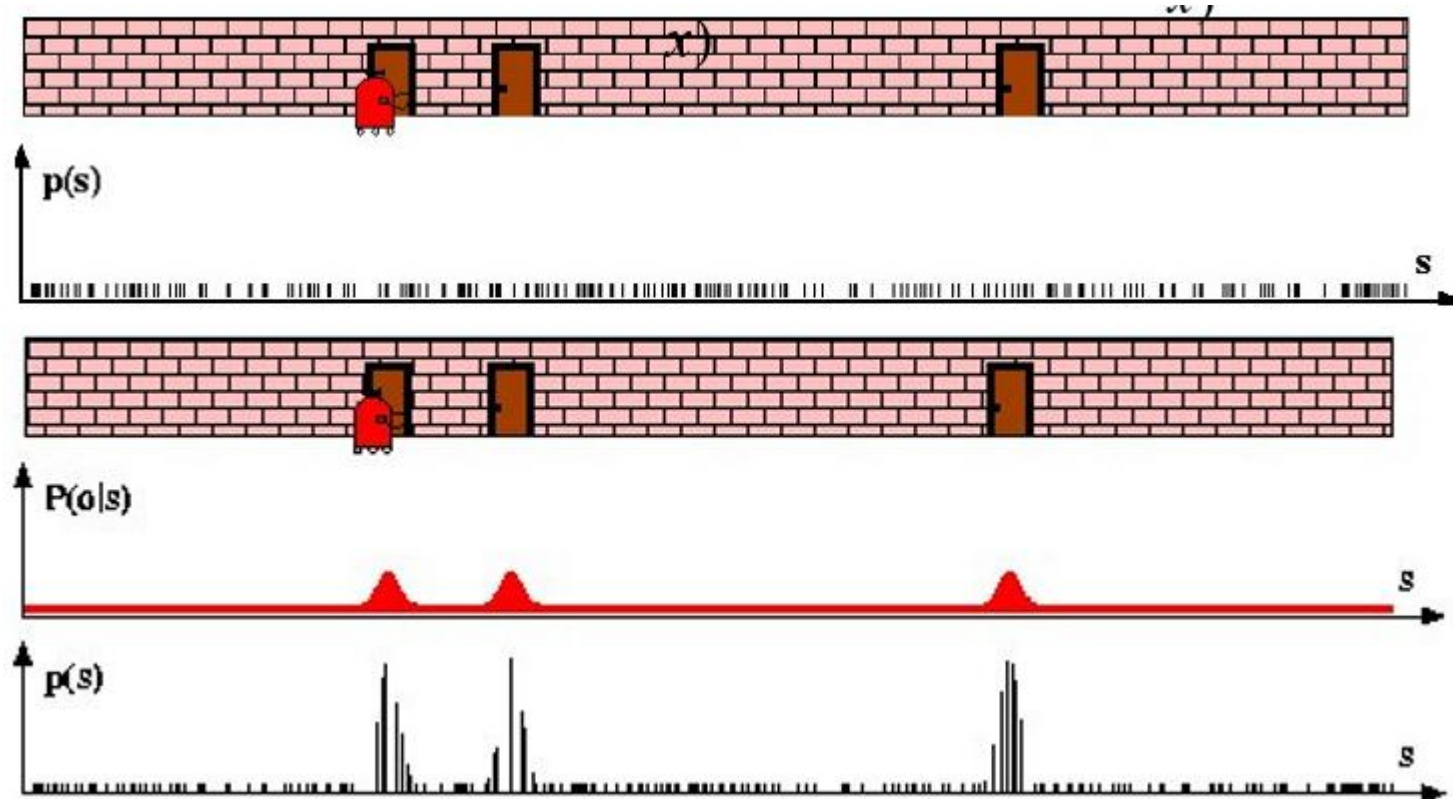


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Sensor Information: Importance Sampling

$$\begin{aligned} Bel(x) &\leftarrow \alpha p(z|x) Bel^-(x) \\ w &\leftarrow \frac{\alpha p(z|x) Bel^-(x)}{Bel^-(x)} = \alpha p(z|x) \end{aligned}$$



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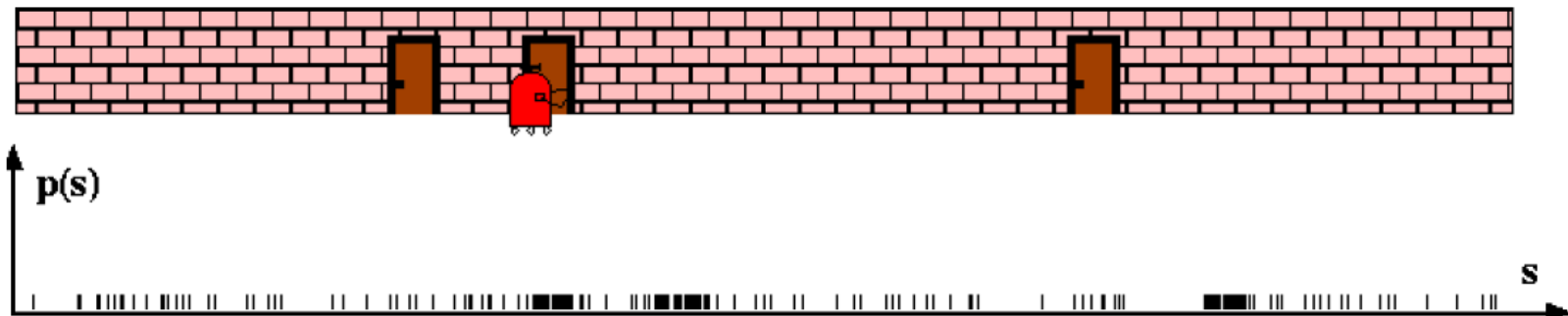
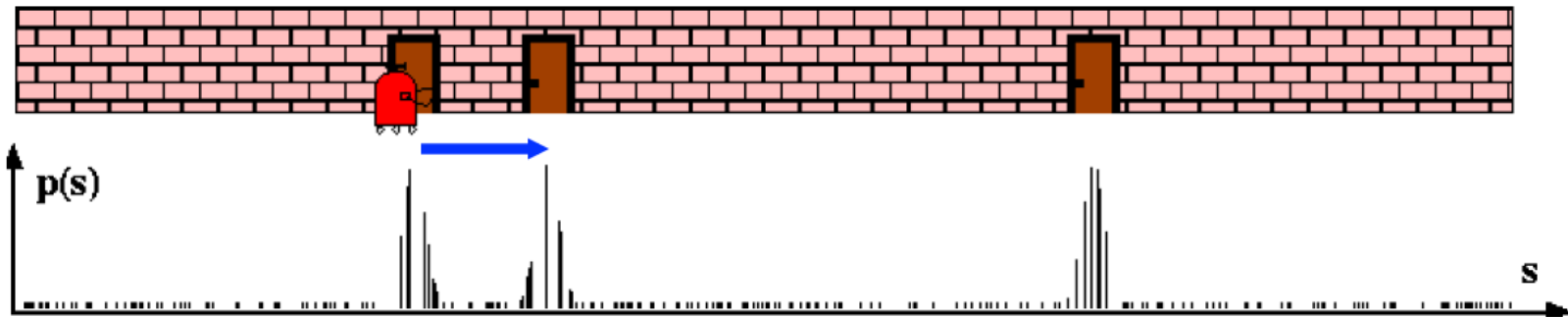


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Robot Motion

$$Bel^-(x) \leftarrow \int p(x|u, x') Bel(x') dx'$$



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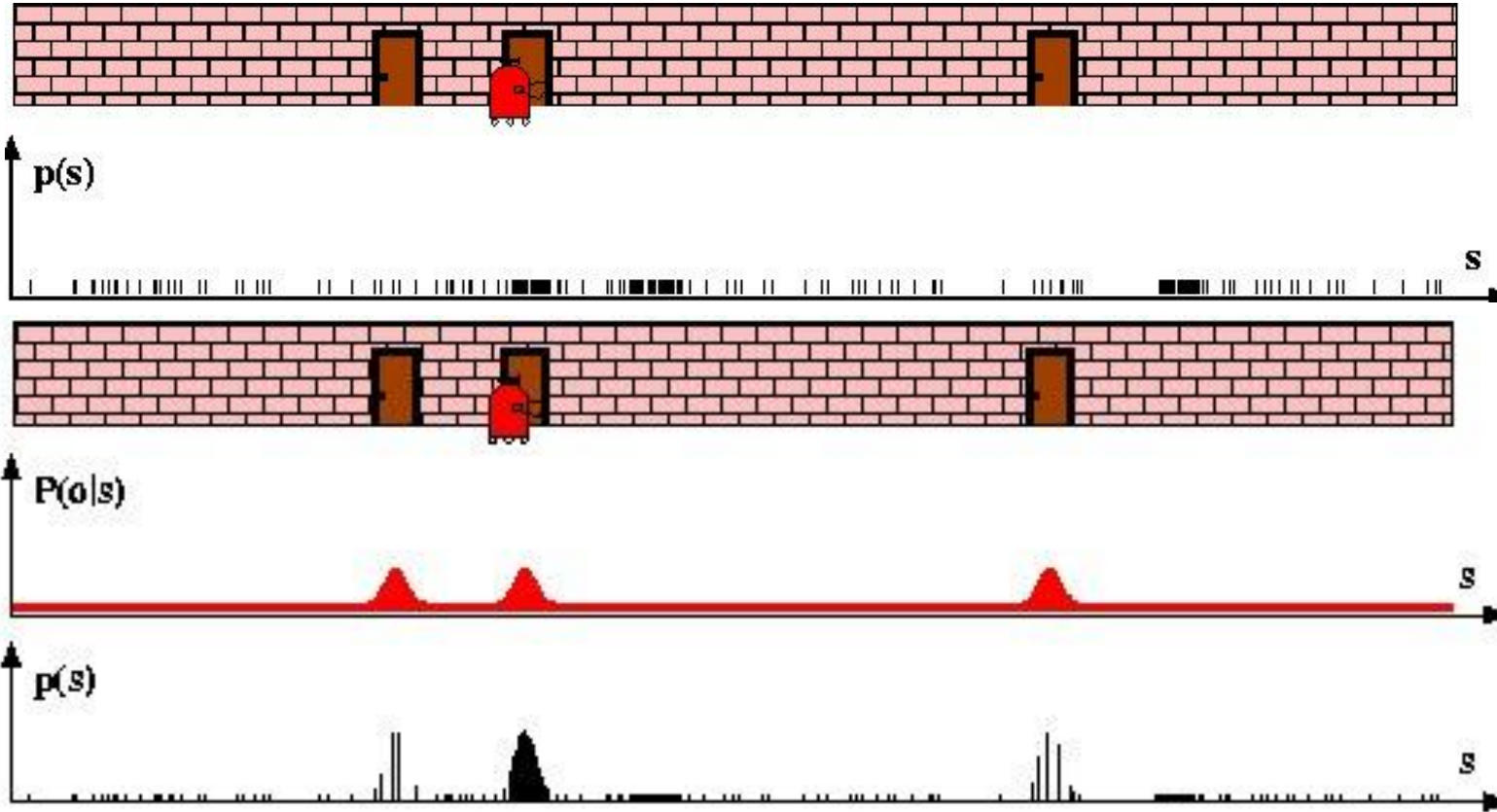


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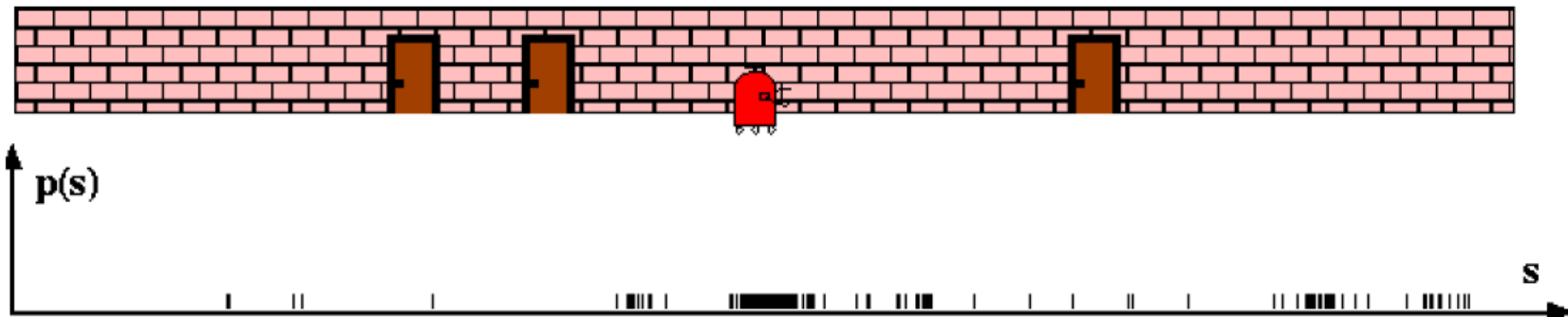
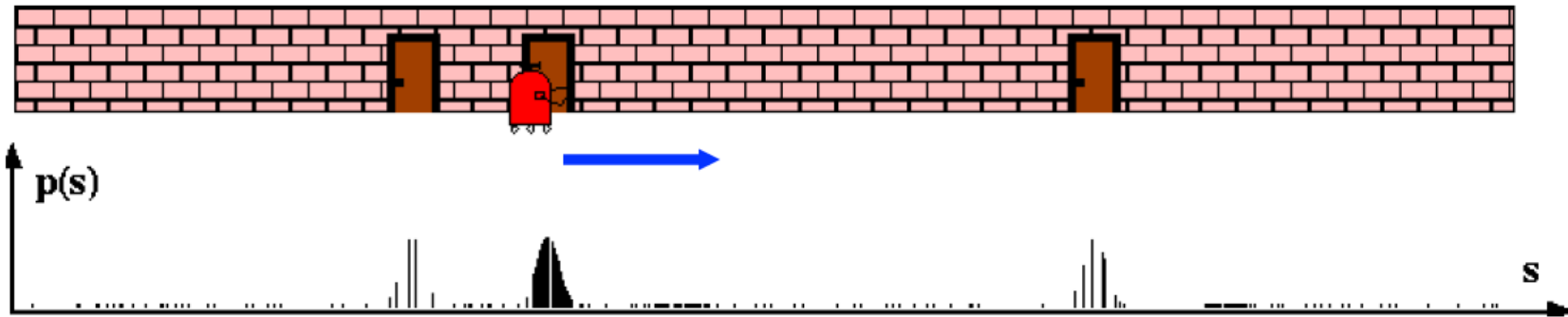


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Robot Motion

$$Bel^-(x) \leftarrow \int p(x|u, x') Bel(x') dx'$$



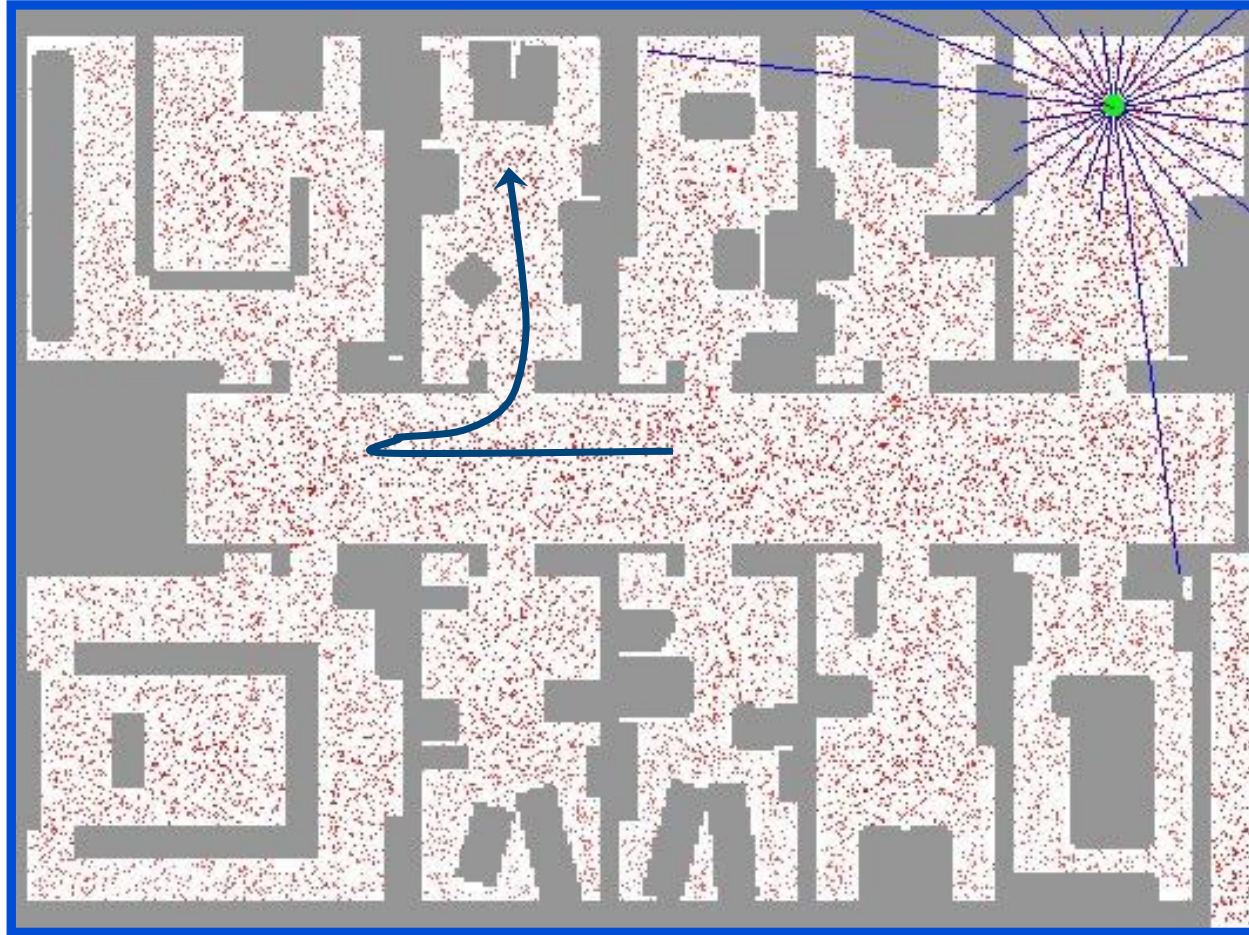
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Monte Carlo Localization



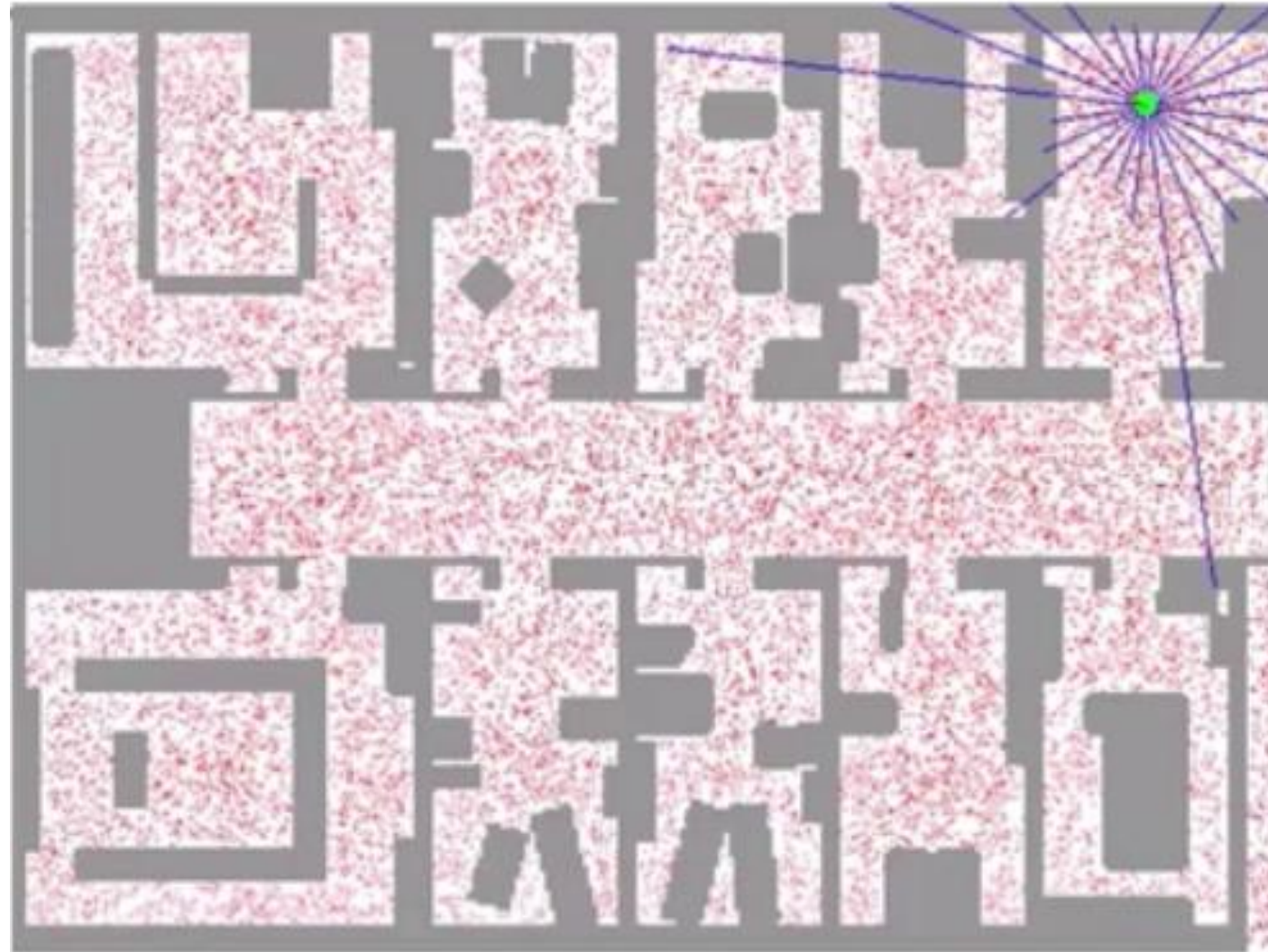
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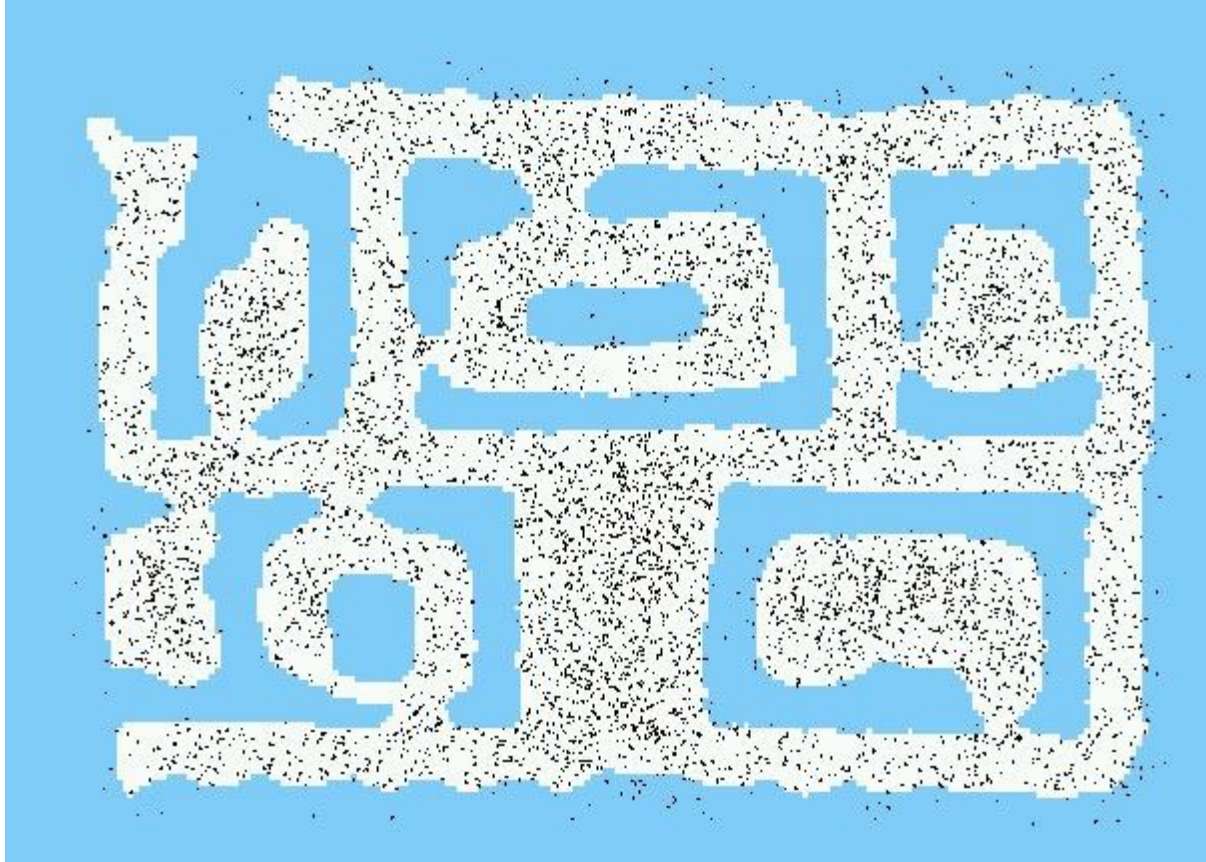
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Initial Distribution



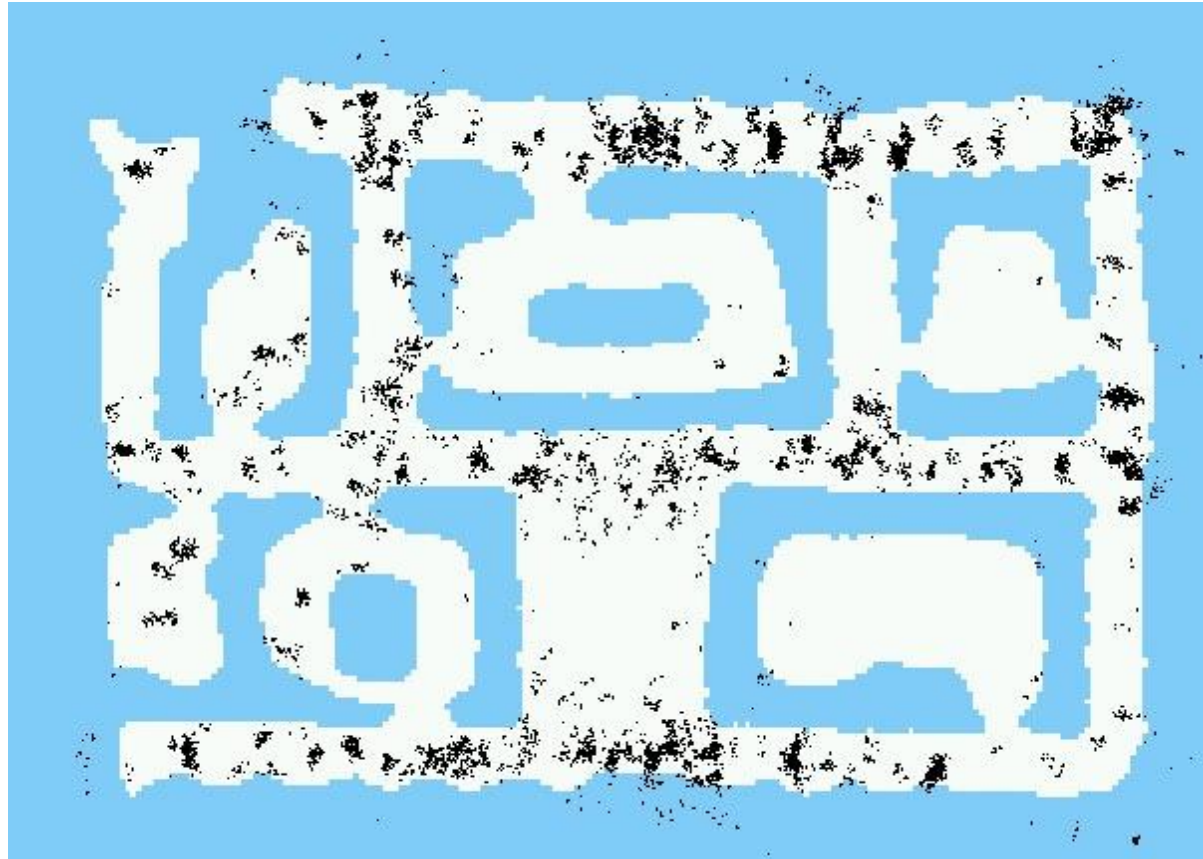
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After incorporating 10 Ultrasound Scans



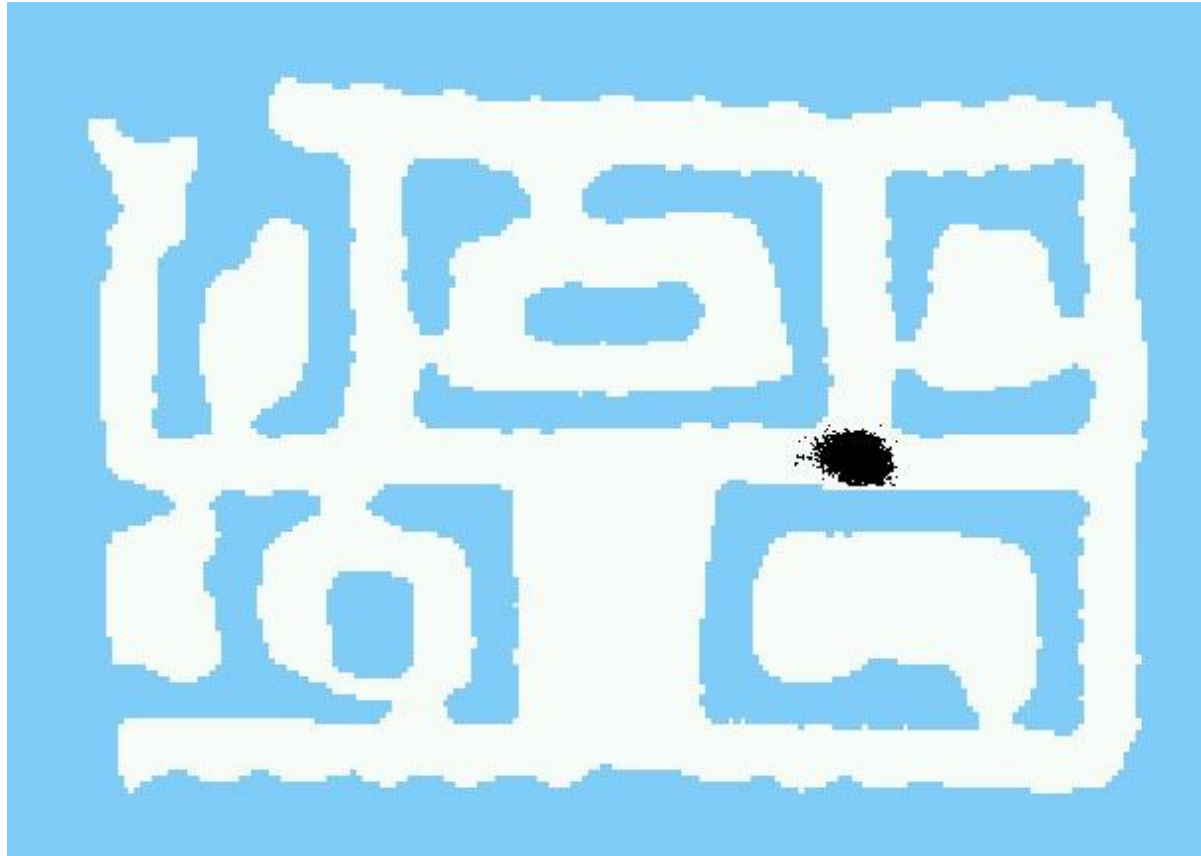
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After incorporating 65 Ultrasound Scans



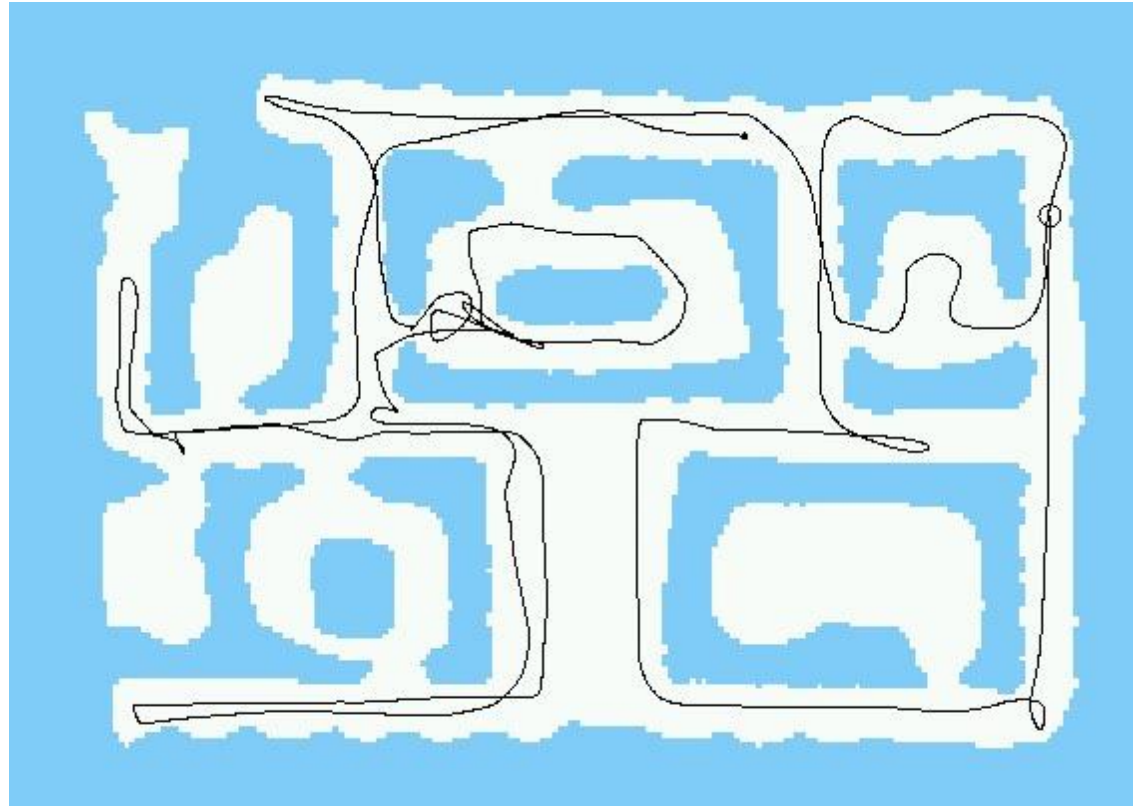
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Estimated Path



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Resampling

- Given: Set S of weighted samples.
- Wanted : Random sample, where the probability of drawing x_i is given by w_i .
- Typically done n times with replacement to generate new sample set S'



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Resampling

- Resampling draw sample i with probability $w_t^{[i]}$ and repeat J times
- Informally: *Replace unlikely samples by more likely ones*
- Survival of the fittest
- *Trick* to avoid that many samples cover unlikely states
- Needed as we have a limited number of samples



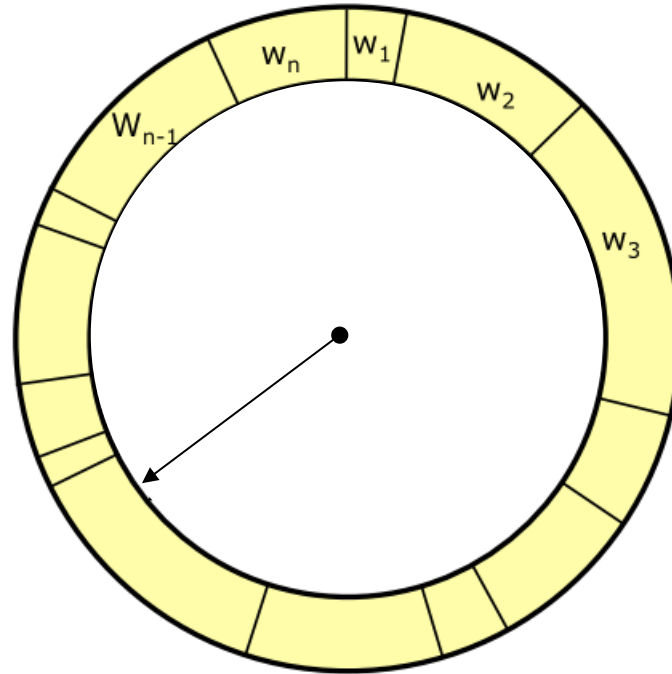
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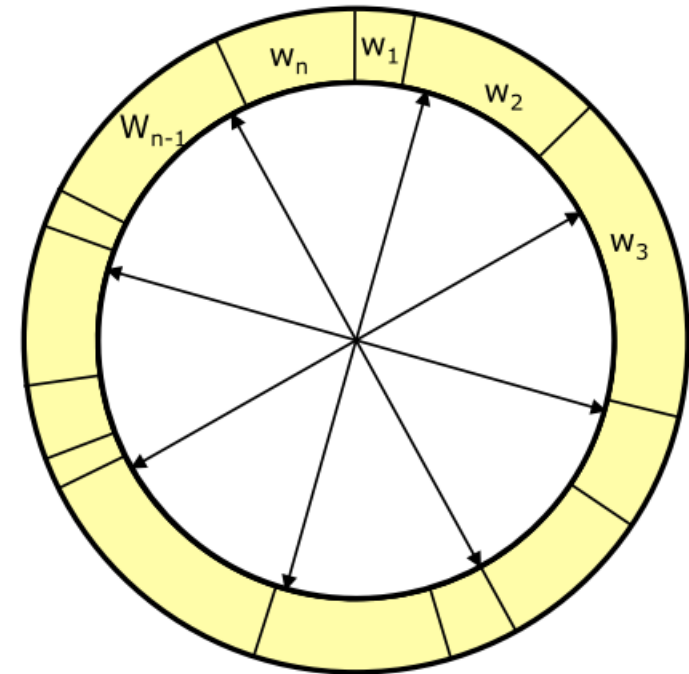


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Resampling



- Roulette wheel
- Binary search, $O(n \log n)$



- Stochastic universal sampling
- Systematic resampling
- Linear time complexity $O(n)$
- Easy to implement, low variance



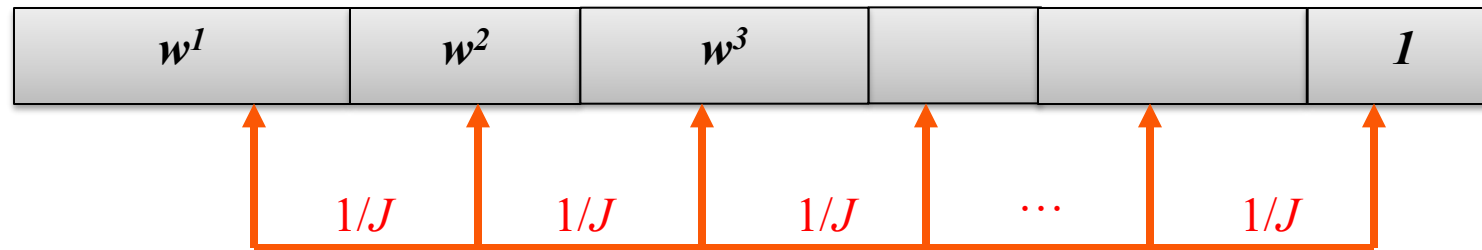
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Resampling



1. Draw random number between **0** and **$1/J$**
2. Pick **$J-1$** particles in **$1/J$** steps



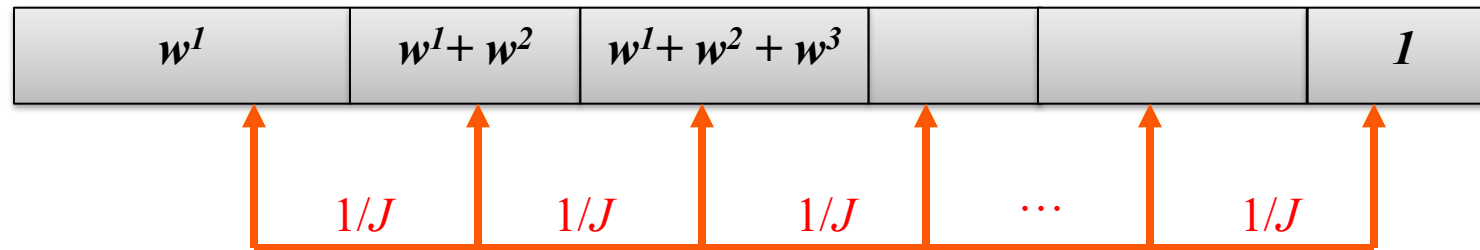
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Resampling



1. Draw random number between **0** and **$1/J$**
2. Pick **$J-1$** particles in **$1/J$** steps
(based on cumulative weights)

for $j = 1 \dots J$

*$U = r + j * 1/J$*

while ($U > \text{cum}[i]$) $i++$

pick_particle i



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Low Variance Resampling

Low_variance_resampling($\mathcal{X}_t, \mathcal{W}_t$):

```
1:    $\bar{\mathcal{X}}_t = \emptyset$ 
2:    $r = \text{rand}(0; J^{-1})$ 
3:    $c = w_t^{[1]}$ 
4:    $i = 1$ 
5:   for  $j = 1$  to  $J$  do
6:      $U = r + (j - 1)J^{-1}$ 
7:     while  $U > c$ 
8:        $i = i + 1$ 
9:        $c = c + w_t^{[i]}$ 
10:    endwhile
11:    add  $x_t^{[i]}$  to  $\bar{\mathcal{X}}_t$ 
12:  endfor
13:  return  $\bar{\mathcal{X}}_t$ 
```

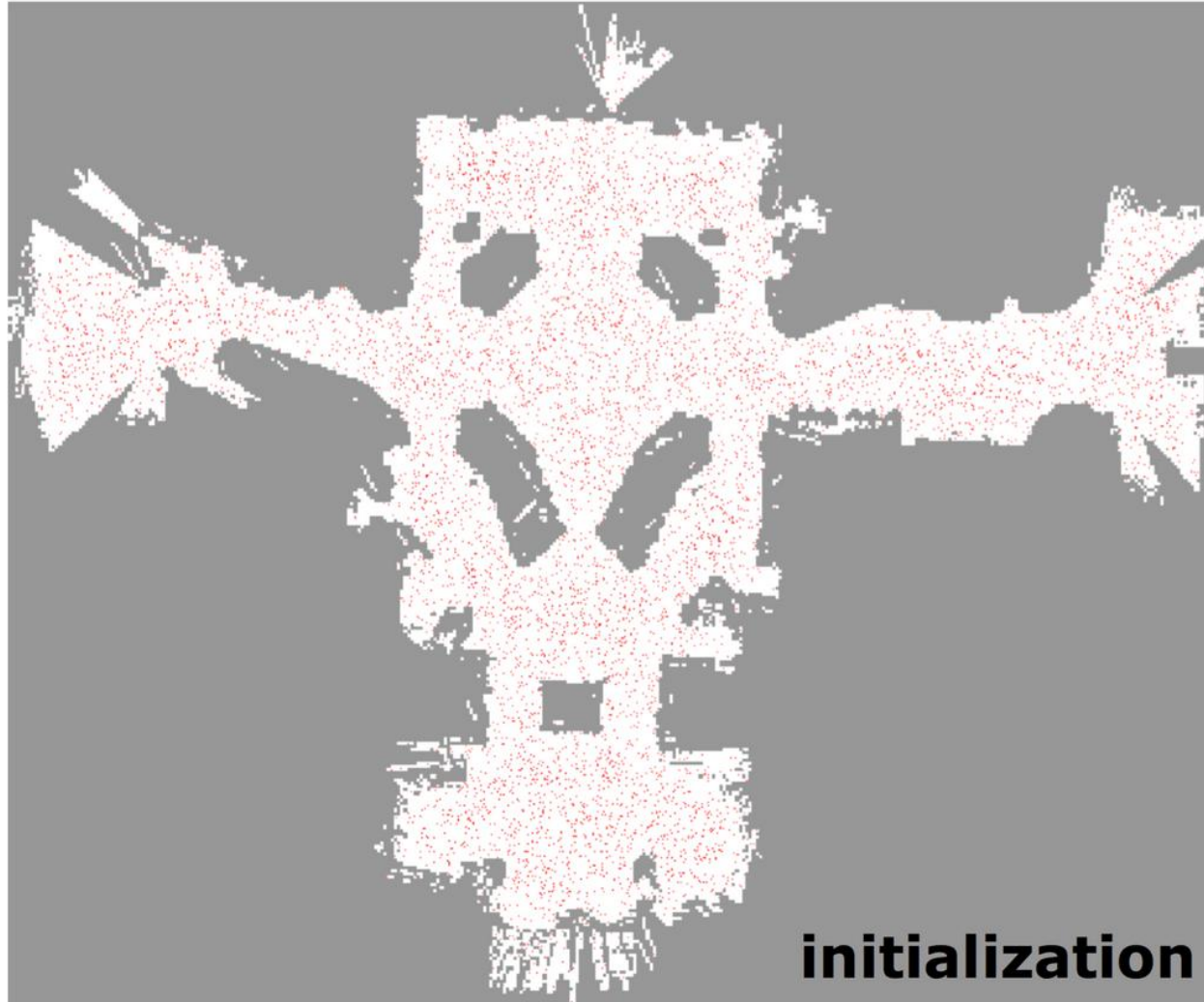




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Particle filter Implementation



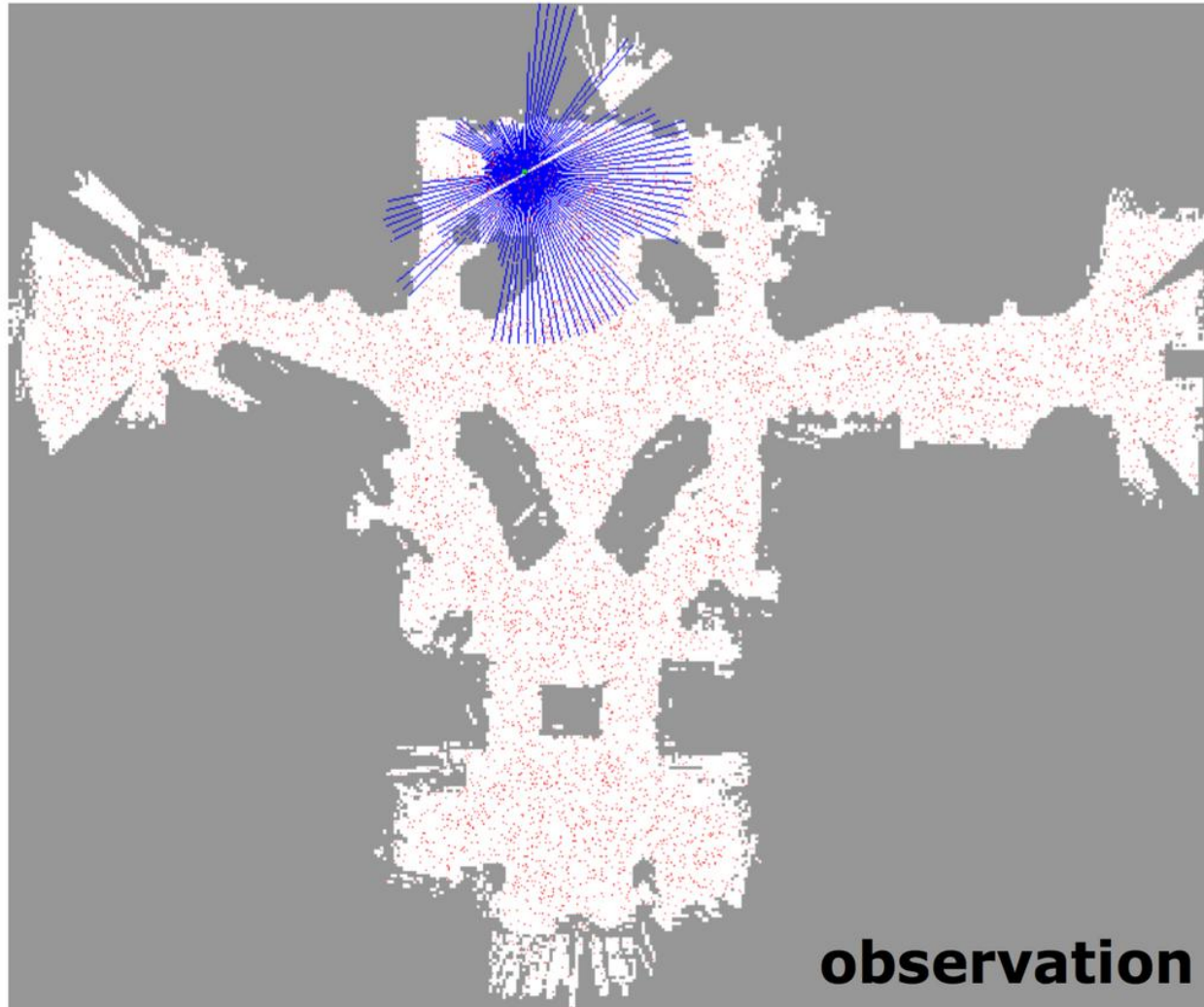
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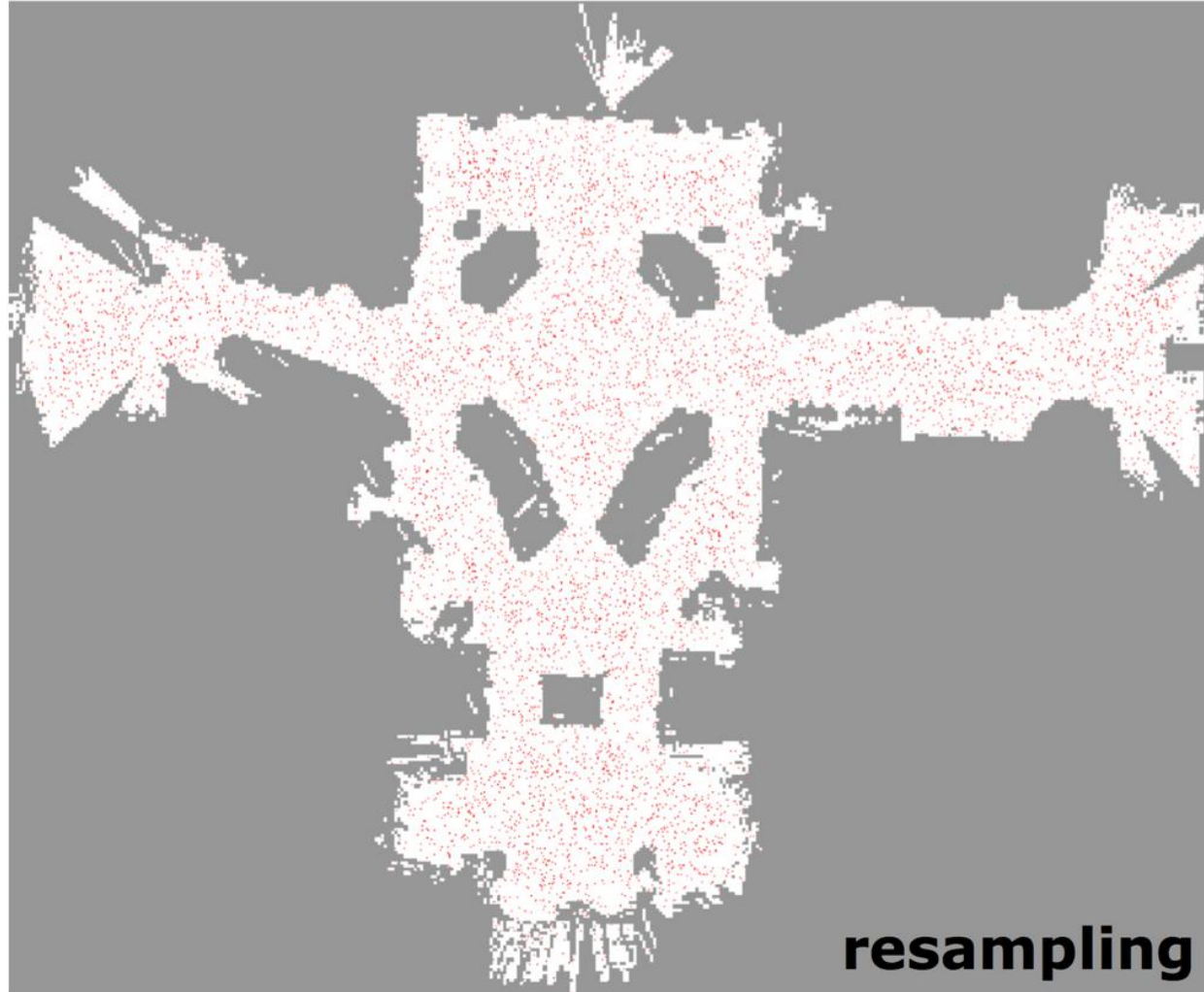
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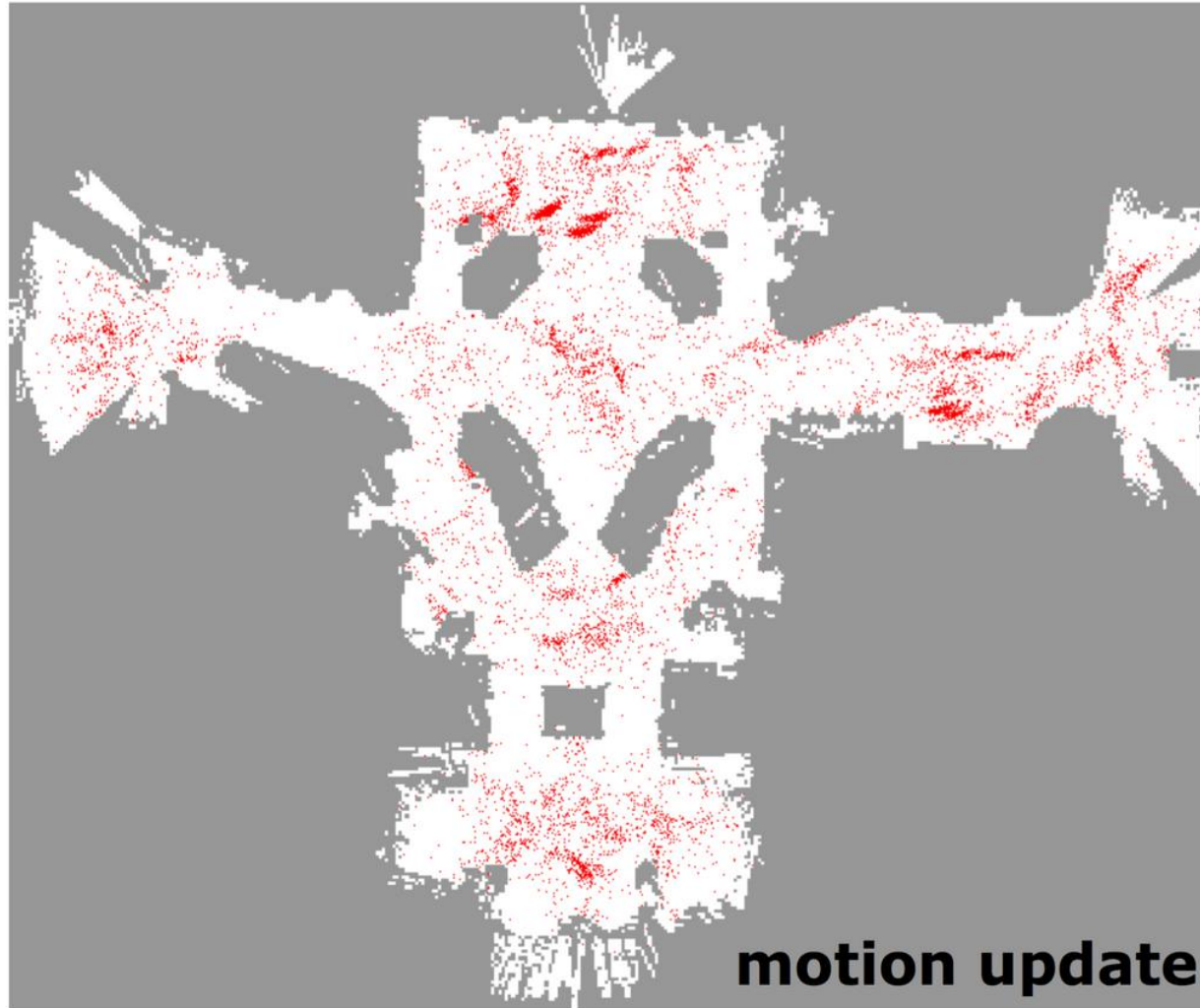
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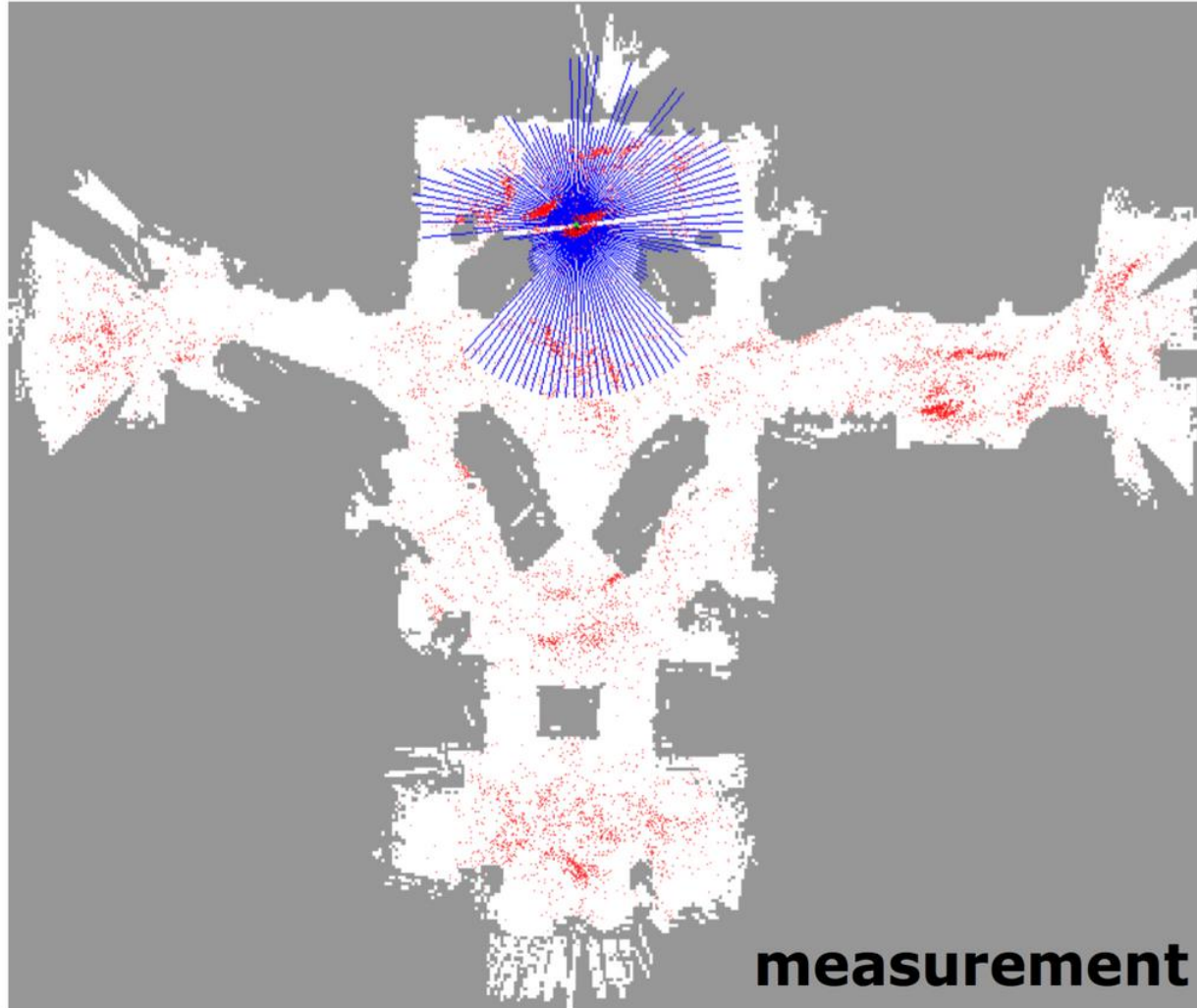
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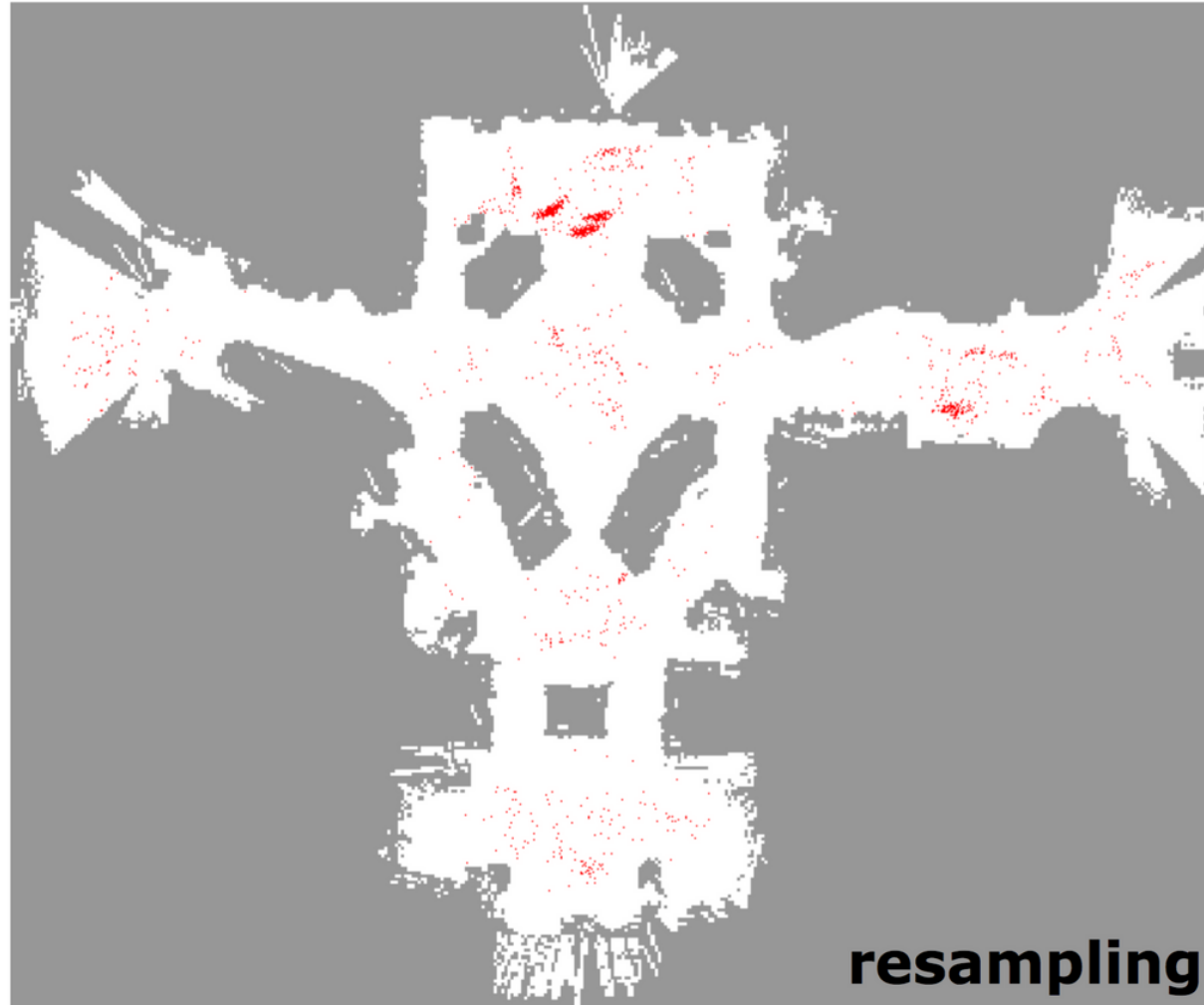
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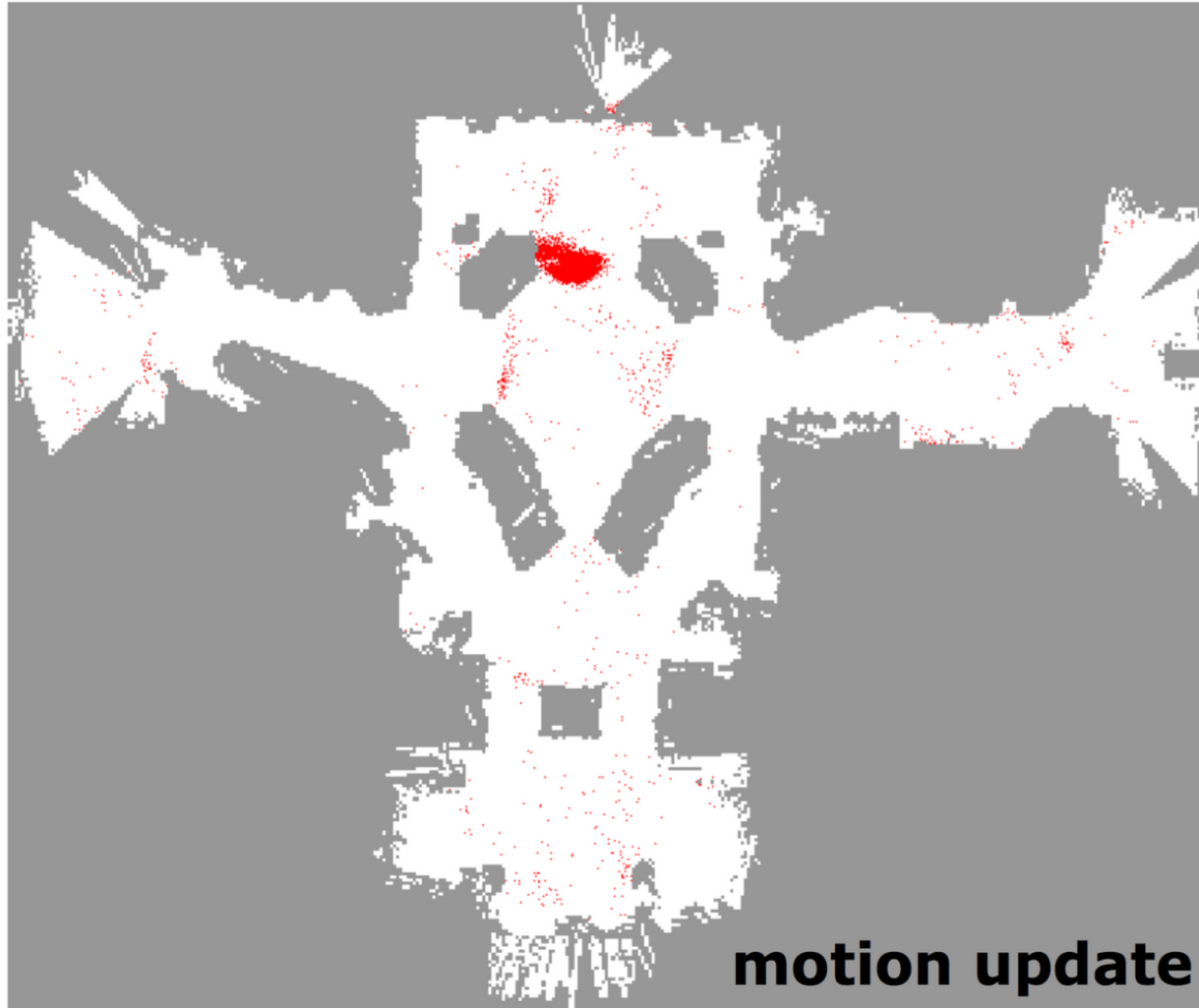
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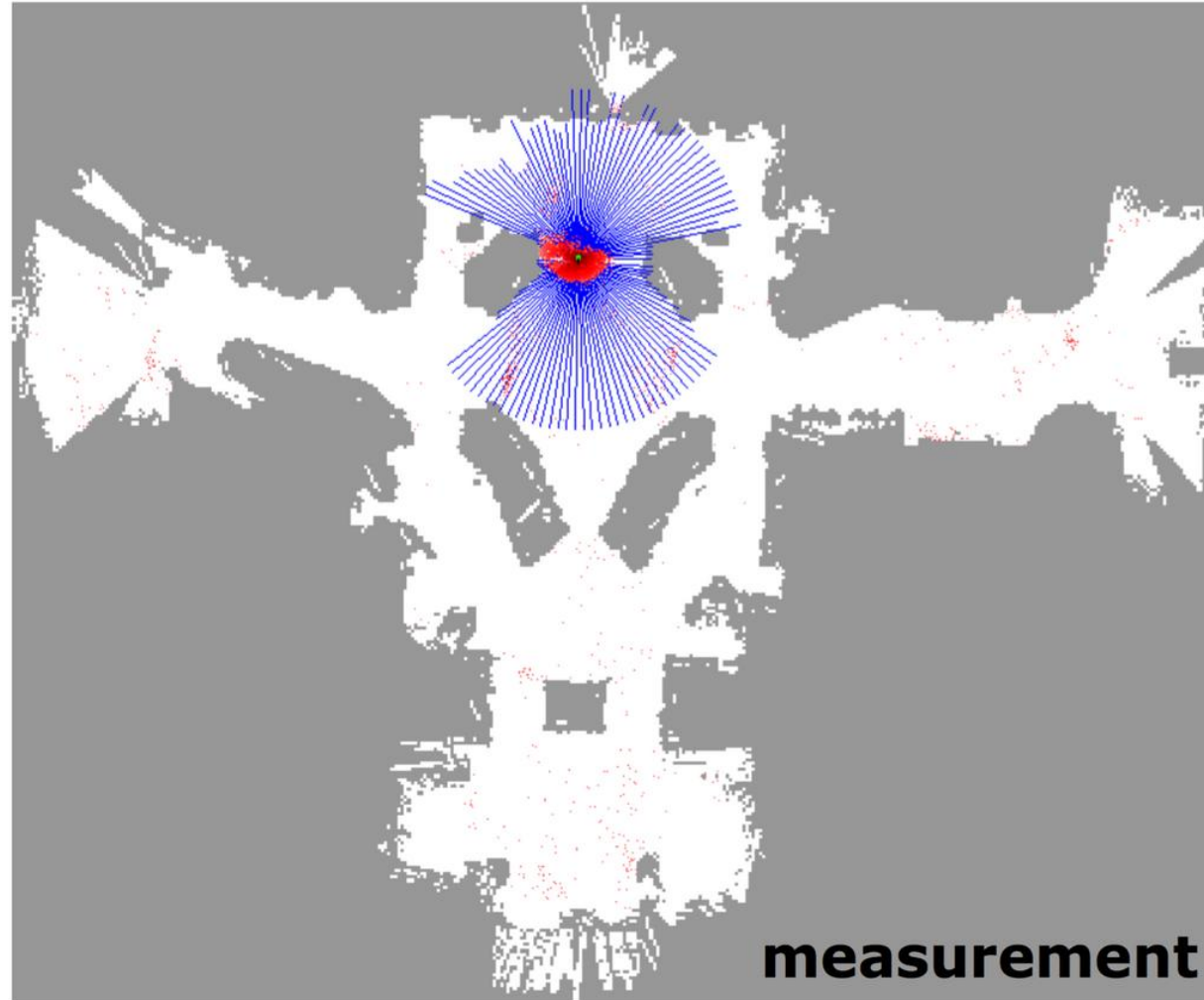
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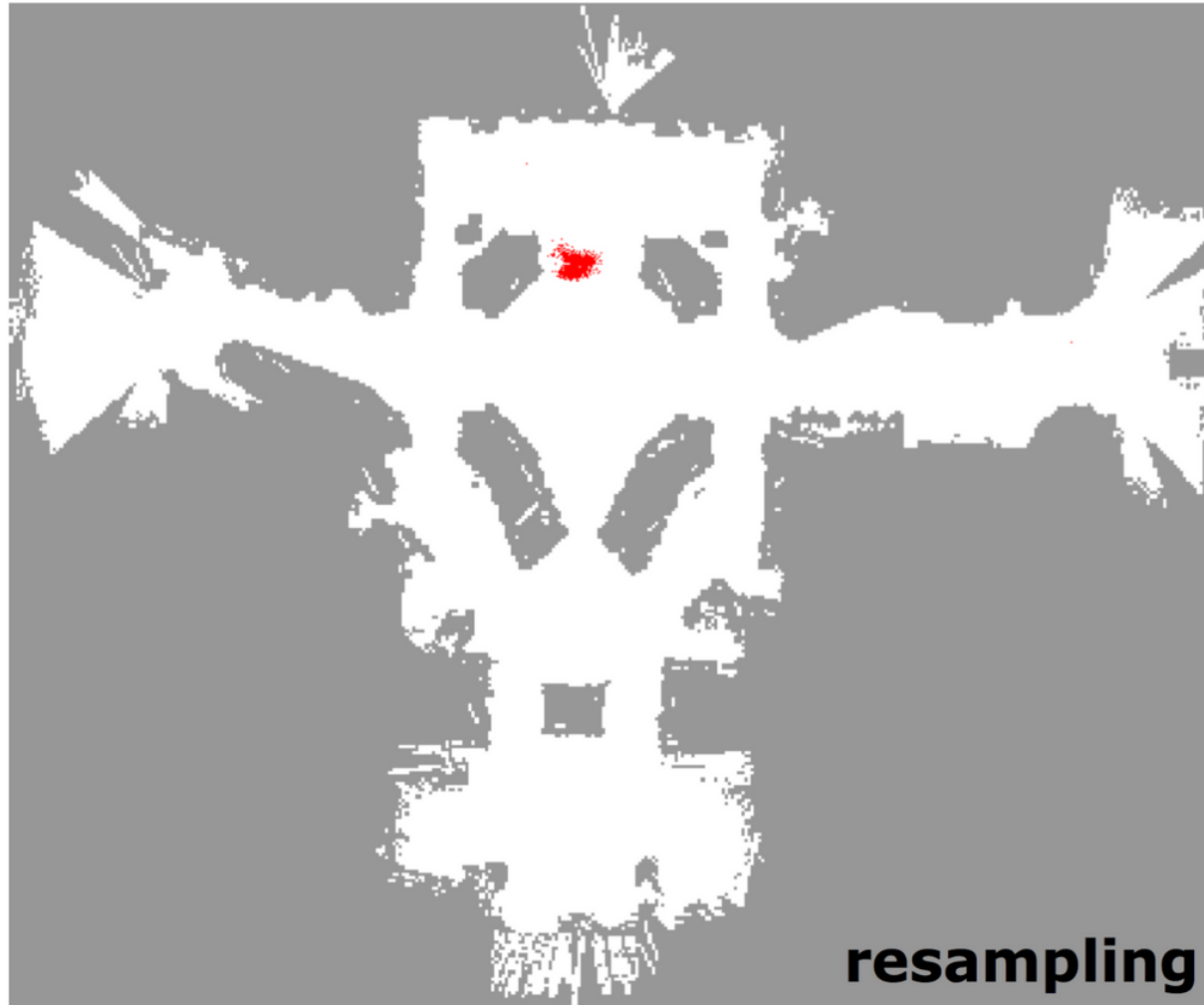
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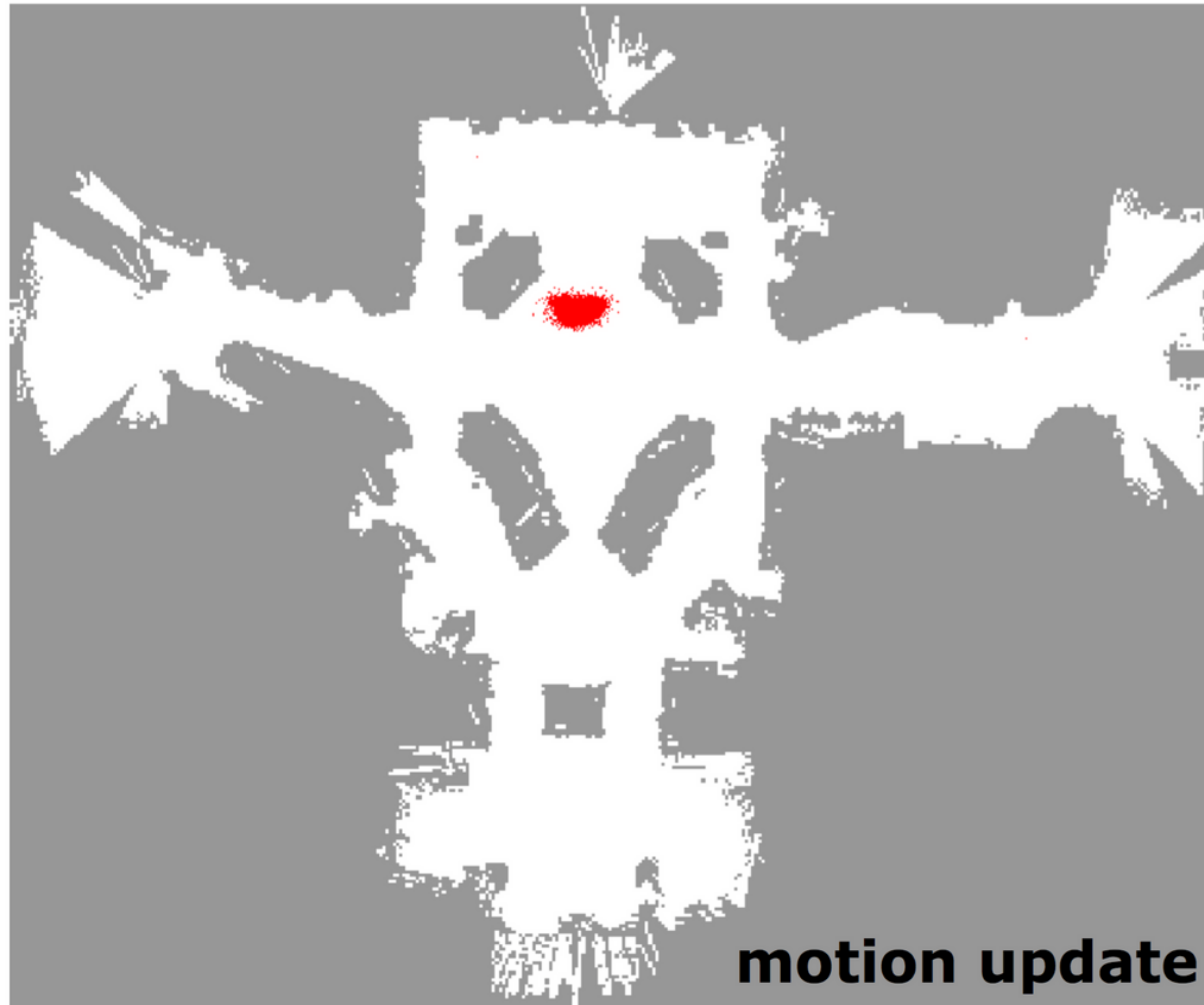
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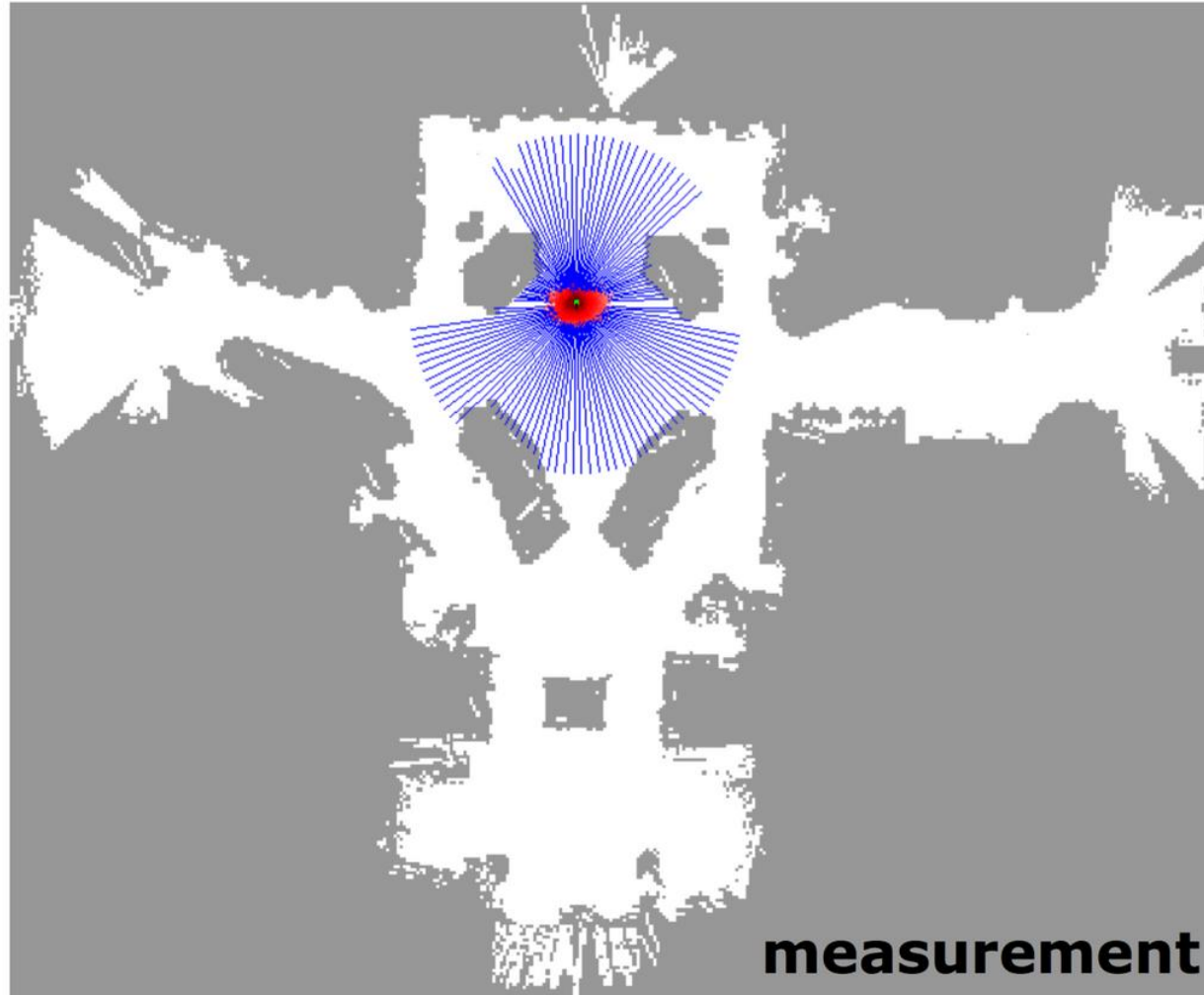
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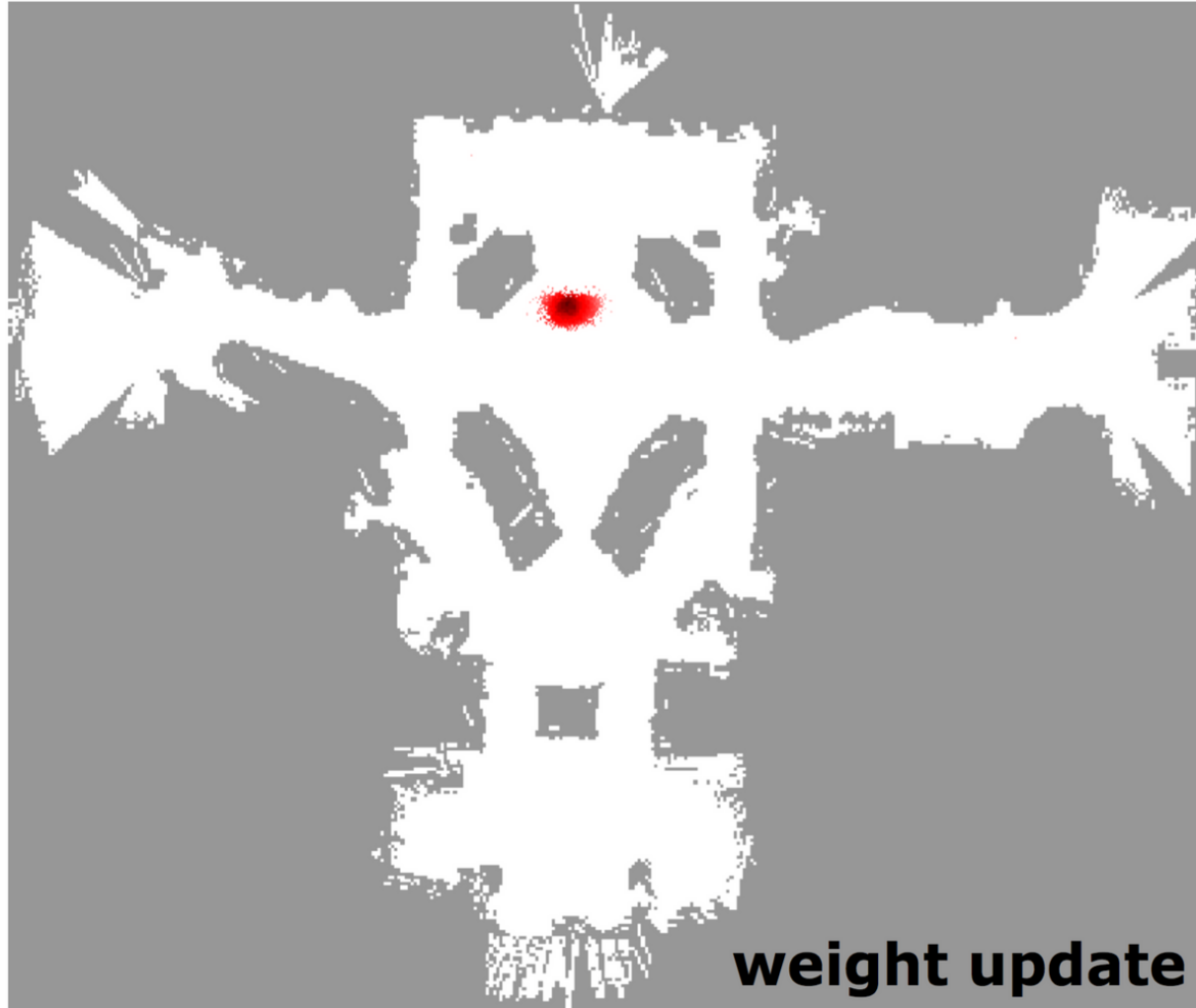
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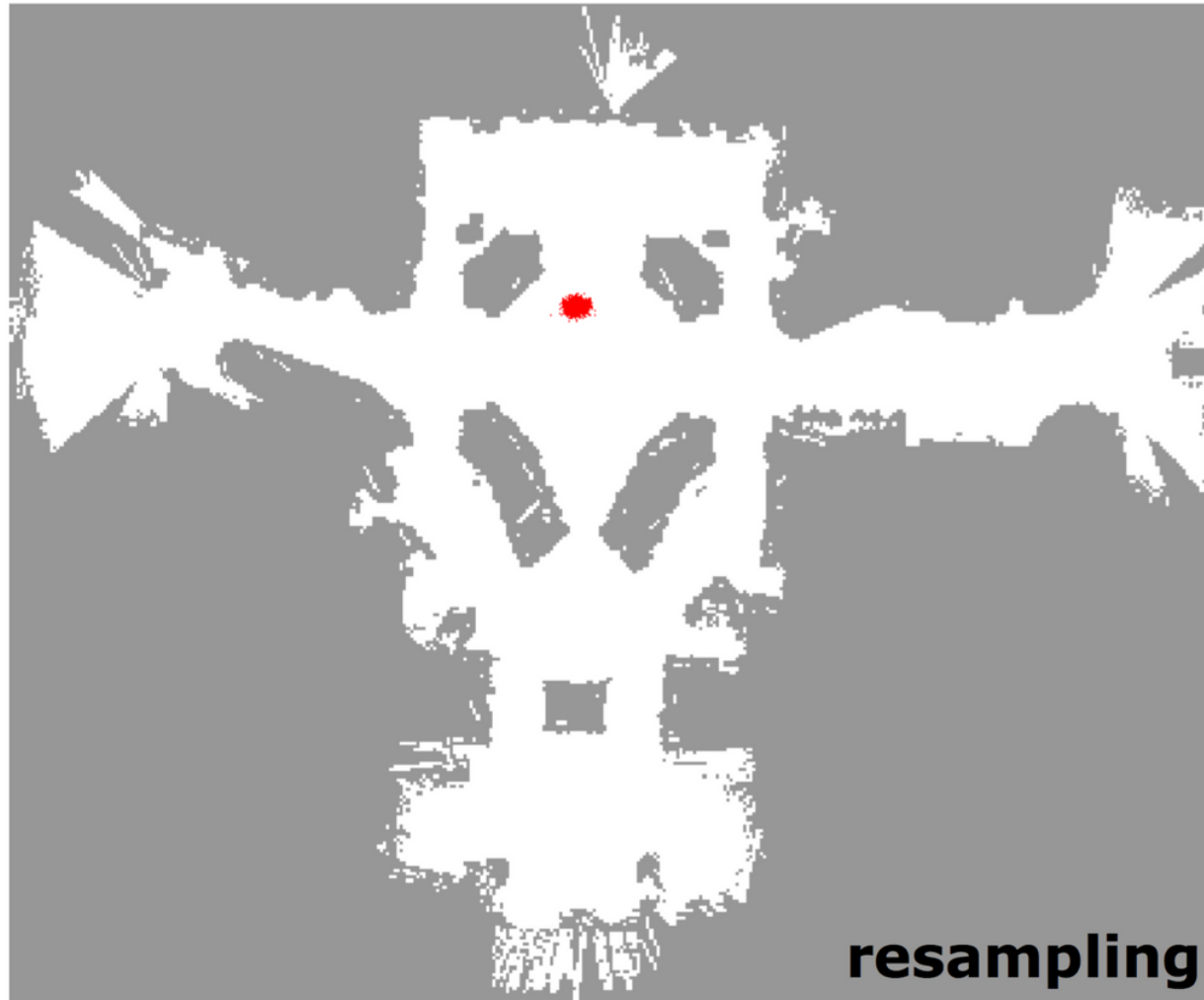
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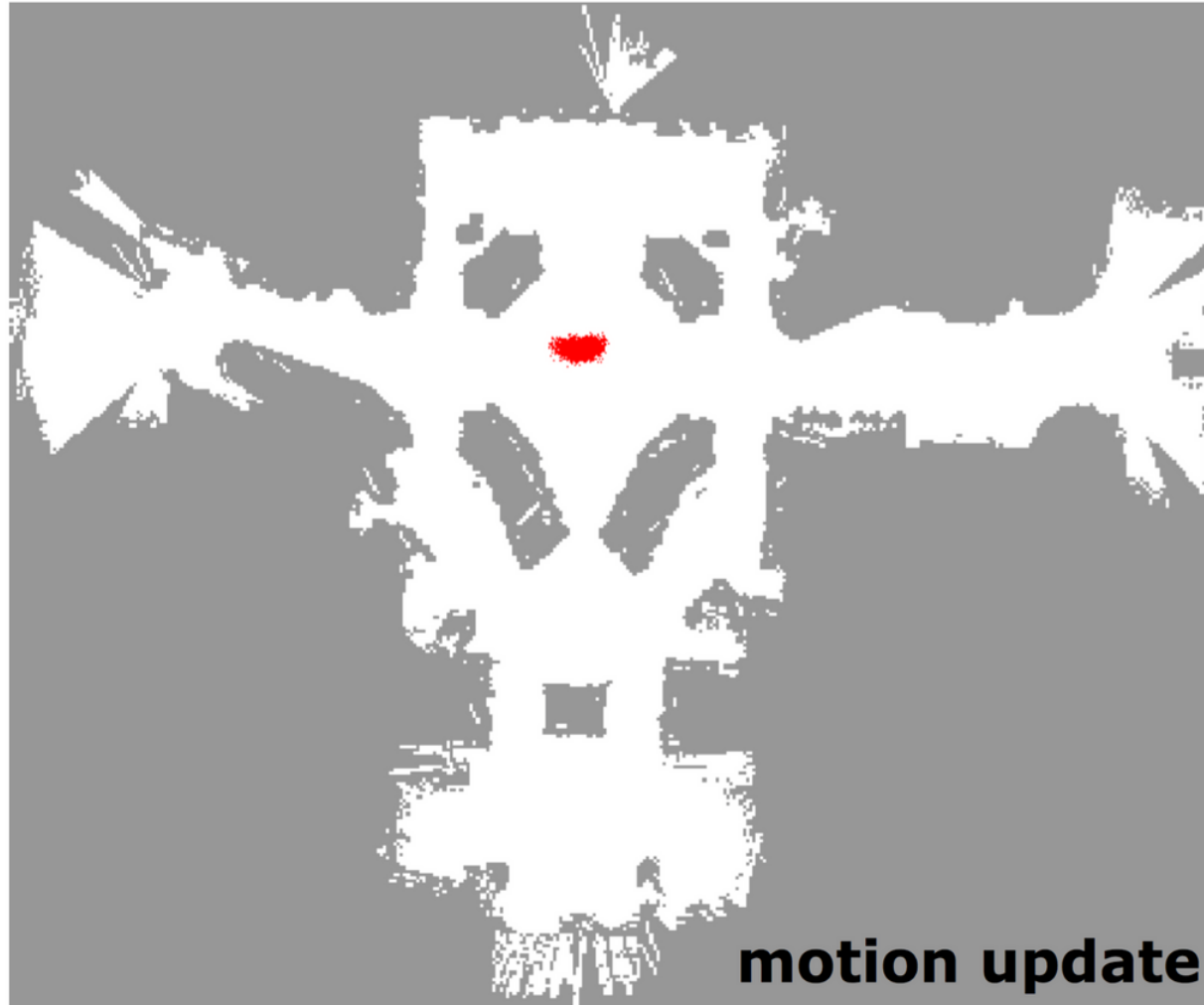
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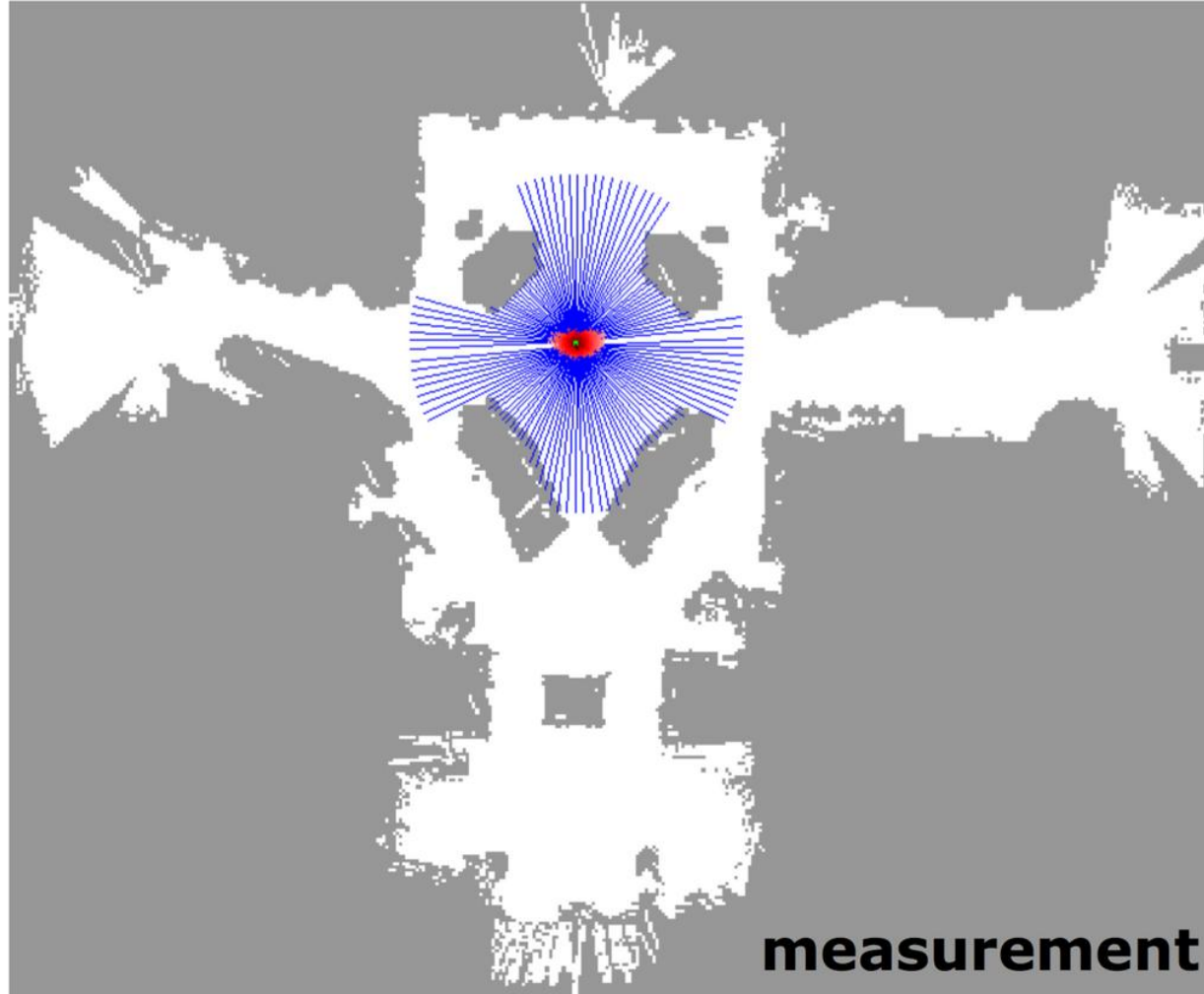
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Summary

- Particle filters are an implementation of recursive Bayesian filtering
- They represent the posterior by a set of weighted samples.
- In the context of localization, the particles are propagated according to the motion model.
- They are then weighted according to the likelihood of the observations.
- In a re-sampling step, new particles are drawn with a probability proportional to the likelihood of the observation.
- Monte Carlo Localization (MCL) is the gold standard for mobile robot localization today



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